Residual Characters of Odd-Degree Fermat Fourfolds

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Abstract

For a Fermat fourfold of odd degree, the rational Hodge classes in middle degree are spanned by character blocks indexed by six-entry Hodge characters. Two direct constructions make such a block algebraic: a contraction of two surface characters, which includes the case of an inverse pair, and an exact 5-standard tuple of Aoki. We call the remaining characters *residual*. We prove that the residual characters at every odd degree are exactly the inflations of six explicit seeds in degrees 33, 39, 45 and 105. The proof is uniform in the degree and uses no computer enumeration. Maximal-order Fourier kernels, a classification of surface characters at odd conductor and an analysis of four points of a five-progression reduce an arbitrary primitive residual conductor to the degrees 15, 21, 33, 39, 45 and 105, which are then settled by the same character equations. An exhaustive census through degree 599, and enumerations at the 103 degrees up to 2001 not covered by the earlier subfamily theorems, give an independent check.

Since the seeds are algebraic, the classification yields a proof of the Hodge conjecture for these fourfolds that does not use Kang's refined motivic dimension estimates. That conclusion is also covered by the written proof of Bhattacharjee [5], and the six seeds were first isolated by Jumagulov [8] through degree 199; the new content is the exact positive classification at every odd degree. We also record the all-dimensional consequences for the prime supports $\{3, 11\}$ and $\{3, 13\}$, and give an explicit family of Fermat characters on which the tensor filtration identity used in Kang's product argument fails.

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1. Introduction

Write

$$X_m^n = \{x_0^m + \cdots + x_{n+1}^m = 0\} \subset \mathbf{P}_{\mathbf{C}}^{n+1}.$$

All cycle groups and cohomology groups in this paper have rational coefficients unless another coefficient field is specified.

Theorem A (positive residual classification). For every odd $m \geq 3$, the orbits of six-entry Hodge characters outside $\mathrm{Dec} \cup \mathrm{Std}_5$ are precisely the inflations of the six orbits in (3.1) whose conductors divide m. Equivalently, the primitive residual orbits consist of one at 33, two at 39, one at 45 and two at 105.

Here Dec means an inverse pair or a contraction of two surface Hodge quadruples, and Std_5 is the exact 5-standard multiset; their definitions are given in Sections 2 and 7. Section 7.8 proves Theorem A without computer enumeration (Proposition 7.21). The residual census of Section 4 and the larger finite checks through 2001 (Corollary 7.13) are independent verifications of the classification, not inputs to its proof.

These predicates describe two direct constructions at the given degree: contraction from surfaces and Aoki's single standard cycle. The theorem identifies exactly when an additional operation is needed beyond these constructions. Section 3 supplies it by a curve-product join, stabilization by inverse pairs, or passage to degree 66. In particular,

$$|\mathcal{R}(m)| = \mathbf{1}_{33|m} + 2\mathbf{1}_{39|m} + \mathbf{1}_{45|m} + 2\mathbf{1}_{105|m}.$$

This counts orbits for the fixed positive criteria; it is not a dimension formula for Aoki's integral gap group.

Corollary 1.1 (algebraicity; compare [8] for $m \leq 199$ and [5]). For every odd integer $m \geq 3$, the map

$$CH^2(X_m^4)_{\mathbf{Q}} \longrightarrow H^4(X_m^4, \mathbf{Q}) \cap H^{2,2}(X_m^4)$$

is surjective. Consequently X_m^4 satisfies the rational Hodge conjecture in every codimension.

The non-middle cohomology of a smooth hypersurface is generated by hyperplane classes. Thus only the displayed map requires proof. The corollary follows from Theorem A and Sections 2–3: the middle ambient class h^2 is algebraic, the primitive Hodge classes are spanned by the rational blocks of Hodge sextuples (Section 2), the two positive criteria and each of the six seed blocks are algebraic, and Lemma 2.1 preserves algebraicity under inflation. It concerns individual fourfolds and does not establish the conjecture for their arbitrary powers.

The following subfamily results organize the proof of Theorem A. Their algebraicity conclusions are also covered by [5].

Theorem 1.2. The rational Hodge conjecture also holds for X_{pq}^4 whenever $p \neq q$ are odd primes, with no upper bound on pq. It is proved in Section 7.

Theorem 7.3 and Proposition 7.4 further establish the exact positive residual classification in squarefree semiprime and prime-power degree. Theorem 7.6 adds every degree coprime to six, without a size bound.

Theorem 1.3. Let $q \geq 7$ be prime and $a, b \geq 1$. The rational Hodge conjecture holds for $X_{3^a q^b}^4$. Moreover its residual orbits outside $\mathrm{Dec} \cup \mathrm{Std}_5$ are exactly the inflations of the degree-33 seed if $q = 11$, of the two degree-39 seeds if $q = 13$, and none otherwise.

Section 7.4 proves the positive classification, including both prime-power depths, directly from primitive Fourier non-vanishing and the six-entry length bound.

Theorem 1.4. The exact six-seed residual equality holds for every odd degree with at most two distinct prime divisors and for every odd degree prime to five. With arbitrary prime support, a primitive residual conductor divisible by nine must be 45 and has the single degree-45 residual orbit.

Sections 7.5–7.7 prove these infinite assertions by classifying small split functions and controlling the lower-order error. In particular the arbitrary-support assertion concerns primitive conductors; inflated residuals may occur at any odd multiple of a seed degree. Section 7.8 completes the unrestricted classification of Theorem A.

Contributions. (i) the exact positive classification of residual characters at every odd conductor (Theorem A and Theorem 7.20), whose six primitive orbits were first isolated in [8] through degree 199, so that the new content is completeness at every odd degree, by a proof that uses no enumeration; (ii) an independent computer check combining the exhaustive residual census through degree 599 with an enumeration of the 103 remaining degrees divisible by 3 through 2001 (Section 4 and Corollary 7.13); (iii) the explicit family on which the tensor identity of Kang's product argument fails (Section 6).

Prior and parallel work. Da Silva [6, Propositions 3.1 and 3.4 in the published version] proved the conjecture for fourfolds of degree at most 100 prime to six and exhibited a degree-33 Hodge class outside the quasi-decomposable and standard criteria considered there. The earlier arXiv version [6, v2, Theorem 3.3] states the algebraicity conclusion for every degree prime to six; that unbounded statement is absent from the published version. Neither version classifies the residual set for the two positive predicates fixed here. Jumagulov [8] proved the conjecture for odd degrees up to 199 and gave the constructions for the six seeds (Theorems 1.2–1.4 there); we do not claim these constructions. His Theorem 1.5 already gives a census through 199. At odd degree his decomposable and quasi-decomposable predicates together coincide with our Dec: a splitting of $a \sqcup \{k, -k\}$ into two Hodge quadruples [8, Lemma 6.1(ii)] either separates $k, -k$, which is a surface contraction with distinguished sum $k \neq 0$, or leaves an inverse pair in a. His standard sextuples are exactly Std_5 [8, Proposition 1.9]. His thirteen orbits beyond these criteria, namely the six seeds and their inflations at 99, 117, 135, 165 and 195, therefore agree with Theorem A through degree 199. In particular the six primitive orbits of (3.1) were first isolated in [8]; the contribution here is the proof that the list is complete at every odd degree. The degree-45 and one degree-105 presentation are instances of Aoki's wider standard lattice calculus [3, Lemma 4.1(iii), p. 185]; [8, Theorem 1.3] gives the explicit identities used below. Peterson [13, Theorem 1.1(C)] gives a proof independent of Kang for all dimensions at degrees $2^a 3^b 5^c 7^d$ not divisible by 420, including 45 and 105. These are prior algebraicity results, not the positive classification for the predicates fixed here.

An earlier manuscript of the author, dated 7 September 2026 and posted on 8 September 2026 [11], stated the all-odd-degree conjecture as a theorem, with a compressed conductor-descent argument. Section 1.1 of [11] also records the failure of the tensor identity of [9, Proposition 2.2(iv)] on an $H^1 \otimes H^1$ diagonal class. Bhattacharjee [5] gives a proof of the Hodge conjecture for X_m^4 at every odd degree by descent along additive-order profiles with terminal characters in degrees 21, 33 and 39; the all-odd statement first appears in its version 1.2.0 of 10 October 2026, and its Remark 3.21 records the same diagonal-class failure.

For $\gcd(m, 6) = 1$ the dichotomy between an inverse pair and an exact 5-standard tuple is [5, Theorem 3.9(ii)], stated there as new when $5 \mid m$ and machine-checked in Lean; Theorem 7.6 reproves it. For $3 \mid m$, [5] abandons classification for an induction on moves (Theorem 3.27: a direct or lowering move, or an exceptional orbit at 21, 33 or 39). The degree-21 orbit lies in Dec; the residual set proved here also contains the second degree-39 orbit and the orbits at 45 and 105, which are not terminal in [5].

Theorem 1.2, the cases of Theorem 1.3, the prime-to-six case (Theorem 7.6) and the all-dimensional consequences for the supports $\{3, 11\}$ and $\{3, 13\}$ (Section 5) are also covered by [5]; the present paper supplies their proofs from the stated classical inputs. The parity detectors of Section 5 are [5, Lemma 4.4], and the short split-function method used in Section 7 is also present in [5].

Favero and Kelly [7, Theorem 1.8 and Theorem 4.8] give a second, independent proof of the Hodge conjecture for the degree-33 Fermat fourfold; the case $m = 33$ is already contained in [8, Theorem 1.1], posted earlier, which [7] does not cite. They pull back classes of cubic scrolls on a cubic fourfold with a symplectic automorphism of order eleven along a monomial rational map [7, (4.3)–(4.4)], and detect the degree-33 character with their residue formula for the primitive Chern character. In the printed equation [7, (4.1)] the last monomial reads x_3^3; the map [7, (4.4)], which sets $y_5 = x_5^{11}$, requires x_5^3. Their method differs from ours: the degree-33 seed here is made algebraic by the level-66 contraction of [8, Theorem 1.4] (Section 3), and the classification of Theorem A concerns every odd degree rather than one character.

Section 6 realizes the tensor-identity failure on the Fermat product $X_m^1 \times X_m^3$ used in [10, (3.3)]. We do not claim the general diagonal-class counterexample as new.

The geometric ingredients are classical. Shioda [16, Theorem II] gives the inductive cohomological structure and its algebraic correspondences; Aoki [2, Theorem 2-1] supplies the standard cycles. The six exceptional orbits used here, their split and two-pair presentations, and their level-66 construction already occur in Jumagulov [8, Theorems 1.2–1.4], whose Theorem 1.5 covers all odd degrees at most 199. Our finite censuses use separately written enumerators and check the classification at every odd degree through 599 and at 103 further degrees through 2001; no proof in this paper depends on them.

Kang [10, Corollary 3.2] states the stronger generalized Hodge conjecture in dimensions three and four in every degree. The proof uses a product estimate from [9, Proposition 2.2(iv)]. Section 6 identifies a false tensor identity in its proof; this does not disprove the product estimate or Kang's final conclusions. Miranda, Movasati, Rufino and Villaflor [12, Theorem 4.4] state an all-dimensional result for degrees $d < 65$, excluding $44, 51, 52$. Their Proposition 4.3 explicitly uses [10, Corollary 3.2]. Corollary 1.1 uses neither of these inputs.

2. Characters and algebraic correspondences

Put $U_m = (\mathbf{Z}/m)^\times$. For a tuple a of nonzero residues whose sum is zero modulo m, define

$$\operatorname{age}_m(ta) = \frac{1}{m}\sum_i [ta_i]_m, \qquad [x]_m \in \{1, \dots, m-1\}.$$

If a has $n+2$ entries, its primitive character space $V_m(a) \subset H^n(X_m^n, \mathbf{C})$ is one-dimensional. Its rational coefficient-Galois orbit is denoted $M_m(a)$. It is of type $(n/2, n/2)$ precisely when

$$\operatorname{age}_m(ta) = (n+2)/2 \quad \text{for all } t \in U_m. \tag{2.1}$$

These assertions are [16, Theorem I, p. 176] and [2, p. 385]. The rational orbit projector belongs to $\mathbf{Q}[G_m]$, where $G_m = \mu_m^{n+2}/\mu_m$. Deck transformations and coordinate permutations are algebraic automorphisms. Therefore one nonzero projection of an algebraic cycle onto $V_m(a)$, together with the rational orbit projector and deck translates, generates $M_m(a)$ by rational algebraic classes. This uses multiplicity one of the irreducible rational deck representation, not rationality of a single complex eigenline. The cycles need not be defined over $\mathbf{Q}$.

2.1. Power maps and non-coprime standard parameters

Lemma 2.1 (inflation and descent). Let $m = kd$, $n \geq 0$, and

$$\pi_k : X_m^n \longrightarrow X_d^n, \qquad [x_i] \longmapsto [x_i^k].$$

This is a finite flat morphism of degree k^{n+1}, and

$$\pi_k^* : M_d(a) \xrightarrow{\sim} M_m(ka). \tag{2.2}$$

The two rational character blocks are algebraic simultaneously.

Proof. On the dense coordinate torus the kernel is μ_k^{n+2}/μ_k, of order k^{n+1}. The power map on projective space is finite, hence its restriction is finite; its image is all of X_d^n. Since the source is Cohen–Macaulay and the target is regular of the same dimension, the finite map is flat. The projection formula gives

$$\pi_{k,*}\pi_k^* = k^{n+1}\,\mathrm{id}\,. \tag{2.3}$$

Equivariance sends the character a to ka. Pullback is injective, and both complex character spaces are one-dimensional. Moreover the reduction map $U_m \to U_d$ is surjective: a unit modulo d can be lifted to avoid the additional primes dividing m, by the Chinese remainder theorem. Thus the entire orbit is the image in (2.2). Pullback preserves algebraic classes. Conversely, for a class in the lower block, push an algebraic representative of its pullback forward and divide by k^{n+1}. This proves both implications. □

The same power-map equivalence is stated, in weighted notation, in [12, Proposition 3.2]; the proof above supplies the implication used here independently of that preprint.

For an odd prime $p \mid m$, put $d = m/p$. Aoki's standard tuple is

$$\sigma_{p,i} = (i, i+d, \dots, i+(p-1)d, -pi), \qquad d/\gcd(i,d) > 2. \tag{2.4}$$

The definition and its admissibility condition are [2, p. 387]. Theorem 2-1 on p. 388 is stated with $\gcd(i,d) = 1$. Since m is odd, d is odd, so $d/\gcd(i,d)$ is odd and exceeds 2 exactly when $d \nmid i$, that is, when $pi \neq 0 \pmod m$. This is the admissibility condition used below whenever a tuple is said to be exactly p-standard.

Lemma 2.2. Every tuple (2.4) has an algebraic rational character block, including the case $\gcd(i,d) > 1$.

Proof. Put $g = \gcd(i,d)$, $m' = m/g$, $d' = d/g$, $i' = i/g$. Then $m' = pd'$, $d' > 2$, $\gcd(i',d') = 1$, and

$$\sigma_{p,i}^{(m)} = g\sigma_{p,i'}^{(m')}. \tag{2.5}$$

Apply [2, Theorem 2-1] at degree m', then Lemma 2.1 in dimension $p-1$. □

Thus the non-coprime case is a proved reduction, not an extra hypothesis silently attributed to Aoki's Theorem 2-1.

2.2. The split structure correspondence

We record the part of Shioda's construction that permits iteration on arbitrary tensors. For a hypersurface of dimension e, the primitive cohomological projector is algebraic; with h the hyperplane class,

$$\Pi_e = \Delta_{X_m^e} - \frac{1}{m}\sum_{i=0}^{e} h^{e-i} \times h^i. \tag{2.6}$$

It removes all ambient cohomology, including the ambient middle line when e is even.

Lemma 2.3 (split join). For any $a, b \geq 0$, there are algebraic correspondences inducing maps

$$J_{a,b} : H^a_{\mathrm{prim}}(X_m^a) \otimes H^b_{\mathrm{prim}}(X_m^b) \longrightarrow H^{a+b+2}_{\mathrm{prim}}(X_m^{a+b+2}), \qquad R_{a,b}J_{a,b} = \mathrm{id}\,. \tag{2.7}$$

Their Hodge types are $(1,1)$ and $(-1,-1)$, respectively. The character of the image is the concatenation of the two characters. No Hodge-class assumption on either input is required.

Proof. In the notation of [16, (2.1), pp. 177–179], set $r = a + 1$, $s = b + 1$, $N = a + b + 2$, and

$$P = X_m^{a+1} \times X_m^{b+1}, \qquad Y = X_m^a \times X_m^b, \qquad Z = \mathrm{Bl}_Y P.$$

Let $j : E \hookrightarrow Z$, $q : E \to Y$ be the exceptional divisor and its projective-line bundle. The morphism $\psi : Z \to X_m^N$ factors as a quotient of degree m, followed by the blowdown in Shioda's diagram. Define, on the indicated primitive summands,

$$J = \Pi_N \psi_* j_* q^*, \qquad R = -\frac{1}{m}(\Pi_a \otimes \Pi_b) q_* j^* \psi^* \Pi_N. \tag{2.8}$$

These maps are algebraic correspondences. We check their composition.

Write $\psi = \bar{\psi} \circ \rho$, where $\rho : Z \to W = Z/\mu_m$ is the finite quotient and $\bar{\psi} : W \to X_m^N$ is the blowup of the two disjoint smooth centres X_m^a and X_m^b. The smoothness of W and this factorization are [17, Lemma 1.4 and Theorem 1.7, pp. 100–102] and [16, (2.1)]. In particular E is the exceptional divisor of $Z \to P$; its image is not contracted by $\bar{\psi}$.

Use the covering deck group $\widetilde{G} = \mu_m^{a+2} \times \mu_m^{b+2}$, acting on the first $a + 2$ and $b + 2$ coordinates of the two factors of P and fixing their last coordinates. It commutes with the quotient action and makes all maps equivariant; its action on the target factors through G_m. Working with this covering group avoids assuming that the target deck group itself lifts to Z. For an input eigenvector z of character $(\alpha; \beta)$, set $x = j_* q^* z$. All entries of α and β are nonzero. The quotient group fixes Y and acts by the same scalar on its two normal directions in P, so it fixes $E = \mathbf{P}(N_{Y/P})$ pointwise. Hence

$$\rho^* \rho_* x = \sum_{g \in \mu_m} g^* x = mx.$$

The blowup decomposition of $H^N(W)$ consists of $\bar{\psi}^* H^N(X_m^N)$ and the exceptional cohomology over the two centres. In every summand over X_m^a, the entire second coordinate block acts trivially: the projective-bundle classes are invariant. Every summand over X_m^b similarly has trivial first block. Thus neither exceptional part contains the character $(\alpha; \beta)$, and

$$\bar{\psi}^* \bar{\psi}_* \rho_* x = \rho_* x, \qquad \psi^* \psi_* x = mx.$$

Also $\psi_* x$ has the same all-nonzero character, so is primitive.

Finally the exceptional normal bundle is $\mathcal{O}_E(-1)$, so, writing $\xi = c_1(\mathcal{O}_E(1))$,

$$q_* j^* j_* q^* z = q_*(-\xi q^* z) = -z.$$

Substitution in (2.8) gives $RJ = \mathrm{id}$. The types follow from the exceptional Gysin map and its retraction. The concatenation rule is [16, (2.12)–(2.14)]. The split join itself was first introduced, in homology, by Ran [14, Section 4, Proposition 4.6 and Corollary 4.7]; see [16, Remark 3, p. 179]. The argument uses the full graded blowup and quotient decompositions, so also applies when N is odd. □

Proposition 2.4 (Shioda's product theorem, with the iteration explicit). Let $r_1, \dots, r_k \geq 0$. If the Hodge conjecture holds for X_m^e for every even

$$e \leq r_1 + \cdots + r_k + 2(k - 1), \tag{2.9}$$

then it holds for $\prod_i X_m^{r_i}$.

Proof. By Künneth, (2.6), and the algebraicity of hyperplane classes, it suffices to treat a Hodge class in a tensor of primitive middle cohomologies, indexed by a subset I of the factors. Let $|I| = s$ and $w = \sum_{i \in I} r_i$. Such a Hodge tensor has w even. For $s = 1$ the assertion is a hypothesis. For $s \geq 2$, compose

$$J_{r_{i_1}, r_{i_2}} \otimes \mathrm{id}, \quad J_{r_{i_1} + r_{i_2} + 2, r_{i_3}} \otimes \mathrm{id}, \quad \dots .$$

This gives an algebraic split injection of type $(s - 1, s - 1)$ into $H^{w+2(s-1)}_{\mathrm{prim}}(X_m^{w+2(s-1)})$. The reverse composition of the R's is an algebraic retraction. Intermediate tensors may have arbitrary Hodge type; Lemma 2.3 applies to their entire cohomology. Only the final image is asserted to be a Hodge class. Its ambient dimension is even and satisfies (2.9), so the hypothesis makes it algebraic. The retraction makes the original tensor algebraic. Reinsert the omitted hyperplane factors to finish. □

For even target dimension, [16, Theorem II(c), p. 178] already states algebraicity in both directions on the second summand, without a Hodge assumption on its input. One may use just that case in Proposition 2.4: join odd-dimensional factors in pairs first, then join the even-dimensional factors. The detailed retraction above also applies to odd target dimension.

This proves the statement of [16, Theorem IV, p. 182] for all k, including $k \geq 3$. The argument does not assume that the first two factors of a Hodge tensor already form a Hodge tensor.

2.3. Surface contractions, curve products and cancellation

Two elementary consequences of [16, Theorem II] will be used. If a Hodge sextuple contains an inverse pair, deleting it leaves a surface Hodge tuple, which is algebraic by Lefschetz $(1,1)$. Joining its block with the corresponding point block gives the sextuple.

If a sextuple splits into triples B, C of sums $k, -k \neq 0$, and $(B, -k)$, (k, C) are surface Hodge tuples, it is algebraic by the first summand of Theorem II, with $r = s = 2$; this is exactly [16, Corollary (ii), p. 179]. The distinguished last coordinates match with opposite signs; the finite-group invariant projector is algebraic. We call these two cases Dec.

Write Std_5 for the sextuples equal as multisets to (2.4) with $p = 5$ and its stated admissibility condition. In particular this means one exact standard tuple, not an integral sum of standard tuples. Repeated entries in all tuples are allowed. Let $\mathcal{H}_6(m)$ be the set of Hodge sextuples and define

$$\mathcal{R}(m) = \big(\mathcal{H}_6(m) \setminus (\mathrm{Dec} \cup \mathrm{Std}_5)\big)/(U_m \times S_6).$$

The content is $g = \gcd(m, a_0, \dots, a_5)$; the primitive conductor is m/g, equivalently the least common multiple of the six additive orders. A tuple is called primitive in the conductor sense when $g = 1$. This usage is distinct from primitive cohomology. For $d \mid m$, inflation is $a \mapsto (m/d)a$. Thus Theorem A asserts the exact equality

$$\mathcal{R}(m) = \{[(m/d)a] : (d, a) \text{ occurs in (3.1)},\ d \mid m\}.$$

If instead the sums of the two triples vanish, the triples are curve characters. The sextuple condition says that their diagonal Galois orbit is of type $(1,1)$ on $X_m^1 \times X_m^1$. Lefschetz $(1,1)$ and $J_{1,1}$ make the sextuple algebraic. This is already [16, Section 4, Lemma 1, p. 183]. In its printed proof the substitution $r = s = 1$ should read $r = s = 2$ in the notation of Theorem II.

Concatenation preserves algebraicity by Lemma 2.3. Inverse pairs can be cancelled by [2, Theorem 1-4(ii), p. 388]. More generally, a known algebraic Hodge factor can be cancelled: apply the algebraic join retraction and contract the resulting product class against an algebraic dual basis of that factor. The intersection form on its primitive Hodge classes is nondegenerate by the Hodge–Riemann relations, so the dual basis has rational algebraic classes. This is [16, Section 4, Lemma 3, p. 183].

3. The six primitive residuals

The six orbits are represented by

$$\begin{array}{r|l} 33 & (1,4,16,22,25,31) \\ 39 & (1,7,16,22,34,37),\ (1,14,16,22,29,35) \\ 45 & (1,19,20,28,30,37) \\ 105 & (1,22,43,64,90,95),\ (3,24,50,66,85,87). \end{array} \tag{3.1}$$

The arithmetic identities below are also checked directly over all units by the accompanying independent Python certificate.

3.1. Degree 33

Put $w = (1,4,16,22,25,31)$, $a = 2w$, at degree 66. Let

$$Q = (1,25,44,62), \quad S = (2,8,32,41,50,65),$$

and let $\beta = (2, 32, 33, 65)$, $\gamma = (8, 33, 41, 50)$. For every $t \in U_{66}$, the ages of Q, β, γ equal two, and those of a, S equal three. The distinguished coordinates 33 of β and γ are opposite modulo 66. Shioda's Corollary (ii) applies at every degree, including even degree; it permits a nonzero distinguished residue equal to its own negative, as here. Their surface contraction therefore gives the algebraicity of S. The equality of multisets

$$a \sqcup (1, 65, 25, 41) = Q \sqcup S \tag{3.2}$$

and inverse-pair cancellation give the algebraicity of a. Lemma 2.1 descends it to w. This uses only the displayed surface blocks, not the Hodge conjecture for every class at degree 66. The same identity and construction are [8, Theorem 1.4].

3.2. Degree 39

The two sextuples have the splits

$$(1, 16, 22) \sqcup (7, 34, 37), \qquad (1, 16, 22) \sqcup (14, 29, 35). \tag{3.3}$$

Each triple has sum zero modulo 39. Every simultaneous unit translate has total age three, so the curve-product criterion in Section 2.3 applies. These are also the degree-39 applications of [8, Theorem 1.2].

3.3. Degrees 45 and 105

At degree 45 the multiset identity is

$$(1, 19, 20, 28, 30, 37) \sqcup (5, 40, 10, 35) = (1, 10, 19, 28, 37, 40) \sqcup (5, 20, 30, 35). \tag{3.4}$$

The first right-hand block is $\sigma_{5,1}$; the second is a surface Hodge tuple (also $\sigma_{3,5}$). At degree 105 the identities are

$$\begin{aligned} (3, 24, 50, 66, 85, 87) \sqcup (15, 90, 45, 60) &= (3, 24, 45, 66, 87, 90) \sqcup (15, 50, 60, 85), \\ (1, 22, 43, 64, 90, 95) \sqcup (85, 20, 100, 5) &= (1, 22, 43, 64, 85, 100) \sqcup (5, 20, 90, 95). \end{aligned} \tag{3.5}$$

The first blocks are $\sigma_{5,3}$ and $\sigma_{5,1}$, respectively. The second blocks are surface Hodge tuples. Each left-hand padding consists of two inverse pairs. Lemma 2.2, surface algebraicity, concatenation, and pair cancellation prove all three assertions. These two-pair presentations are included in [8, Theorem 1.3].

For clarity, the non-coprime standard parameters appearing in these identities reduce as follows:

$$(m, p, i, g, m/g) = (45, 3, 5, 5, 9), \quad (105, 5, 3, 3, 35), \quad (105, 3, 15, 5, 21). \tag{3.6}$$

In each reduced case $\gcd(i/g, (m/p)/g) = 1$, exactly as required by [2, Theorem 2-1].

4. Finite censuses

Proposition 4.1 (computer census through degree 105). At the 52 odd degrees from 3 through 105 there are, in total, 57 pair-free Hodge orbits. Of these, 15 are exactly 5-standard and 35 of the remaining orbits admit a surface contraction. The seven residual orbits are the six primitive orbits in (3.1) and the degree-33 inflation at 99.

Proof. Both independent enumerators described in the Appendix check all 52 degrees, including those covered by Aoki's theorem. The orbit normalization and exhaustive loop bounds are justified below. Each candidate is checked at every unit; classification tests all ten three-three splits with nonzero distinguished sum and all admissible standard parameters. The implementations use different enumeration algorithms and separate closure predicates. Rebuilding and running both gives equal canonical orbit sets in each of the three disjoint categories at every degree, with the totals just stated. Every raw candidate from the direct enumerator and every classified pair-free orbit from the three-plus-three enumerator is retained in Supplement S1. No conductor reduction or expected seed list restricts either enumeration. □

The direct implementation enumerates the divisor-normalized candidates by four nested loops, followed by the forced sixth entry and all unit age equations. Its Python classification uses the definitions of Section 2.3, with all ten surface splits and every admissible standard parameter. A fresh replay at all 52 degrees agrees with a fresh three-plus-three

enumeration in every pair-free orbit and its classification, as well as all standard, decomposable and residual counts; the candidate rows and the replay entry point are in Supplement S1. The proof of Theorem A in Section 7 does not use this census (Proposition 7.21); it and the extensions through 599 and 2001 are independent checks.

The content of a sextuple is $\gcd(m, a_0, \dots, a_5)$. If this is $g > 1$, division by g gives a Hodge sextuple at degree m/g, by the surjectivity of the unit reduction map. It suffices to classify content-one tuples and then use Lemma 2.1.

When $\gcd(m, 30) = 1$, [1, Theorem A, p. 24] says that every Hodge sextuple is a union of inverse pairs: every prime divisor of m exceeds six. Thus 159 of the 299 odd degrees between 3 and 599 require no enumeration. Aoki's 1984 erratum, p. 572, replaces the statement of Theorem B on p. 24; it does not change Theorem A or the character nonvanishing and generation results used here.

For each of the remaining 140 degrees, the program `primitive_residual.c` enumerates all content-one Hodge sextuples outside Dec and the exact family (2.4) for $p = 5$. The enumeration has the following completeness proof. For any nonzero coordinate x, put $g = \gcd(x, m)$. There is a unit t such that $tx = g \pmod m$: solve $t(x/g) = 1 \pmod{m/g}$, and use unit surjectivity to lift t modulo m. Therefore each orbit has a representative with its first coordinate a proper positive divisor g of m. After fixing that coordinate, sort the remaining five. Enumerate their first four entries, and recover the fifth from $\sum_i a_i = 3m$. The loop bounds impose only this sum and this sorting condition, so no Hodge representative is excluded. For each candidate, the program checks (2.1) at every unit. It then checks the precise surface-splitting and standard criteria, rejects nonprimitive content, and takes the integer lexicographic minimum under the unit action.

The full output consists of one completion row for each of these 140 degrees, with the six residual rows (3.1). The largest tested degree with $3 \mid m$ or $5 \mid m$ is 597. The run used GCC 11.5.0 on Linux, four processes, and approximately 107 seconds of elapsed time. Its completion rows, the complete source, and a separate exact arithmetic checker are supplied with this paper.

Inflation preserves and reflects Dec. An inverse pair divides to an inverse pair. For a surface split, its distinguished residue is the sum of three original coordinates, so is divisible by the content g; division gives the lower-degree split and preserves every age equation. The forward direction is the same calculation. Inflation also preserves and reflects the exact 5-standard family. In the converse direction, a standard progression of step $m/5$ all of whose coordinates are divisible by g requires $g \mid m/5$, so $5 \mid m/g$, and divides to a standard progression at degree m/g. Consequently the full residual set through 599 consists exactly of the 38 odd inflations of (3.1). In the following proof the exhaustive input is the primitive census, not this count.

Algebraicity through degree 599, independently of Theorem A. If $\gcd(m, 30) = 1$, every Hodge sextuple is a union of inverse pairs [1, Theorem A], hence is generated by linear subspaces [2, Theorem 1-1]. In every other degree, reduce the content. A primitive tuple is in Dec, is a 5-standard tuple, or is in (3.1). Sections 2 and 3 prove algebraicity in each case. Pull back to the original degree and use deck transformations and coordinate permutations to cover all ordered rational character blocks. Add the ambient class h^2. □

5. All dimensions and products for two prime-support families

Theorem 5.1. Fix $q \in \{11, 13\}$. If $m = 3^a q^b \geq 3$, $a, b \geq 0$, then the rational Hodge conjecture holds for X_m^n in every dimension. It also holds for every finite product $\prod_i X_{m_i}^{n_i}$ with all m_i in this fixed prime-support family.

Proof. For a tuple T at degree m, write $\nu(T) = (n_x)_{x \neq 0}$ for its vector of multiplicities. Let B_m be Aoki's integral Hodge group of such multiplicity vectors, and $S_m \subseteq B_m$ the subgroup generated by inverse pairs and standards. The gap-group order in [4, Theorem 3.1, p. 107], which [4] derives from Yamamoto's theorem and [1, Theorem D], gives $B_m/S_m \cong \mathbf{Z}/2$ at degrees 33 and 39. Define the parity detectors on multiplicity vectors $n = (n_x)$,

$$\lambda_{33}(n) = n_1 + n_{10} + n_{23} + n_{32}, \qquad \lambda_{39}(n) = n_1 + n_{14} + n_{25} + n_{38} \pmod 2. \tag{5.1}$$

Their support is the unit set $D_{3q} = \{x \in U_{3q} : x \equiv \pm 1 \pmod q\}$, which satisfies $-D_{3q} = D_{3q}$; hence every inverse pair meets D_{3q} in 0 or 2 entries. In an admissible $\sigma_{3,i}$, the entry $-3i$ is a nonunit, and the progression $i, i+q, i+2q$ is constant modulo q while its residues modulo 3 run through $\mathbf{Z}/3\mathbf{Z}$; it meets D_{3q} in two entries if $i \equiv \pm 1 \pmod q$ and in none otherwise. In an admissible $\sigma_{q,i}$, so $3 \nmid i$, the entry $-qi$ is a nonunit, and the progression $i, i+3, \dots, i+3(q-1)$ consists of units modulo 3 whose residues modulo q run through $\mathbf{Z}/q\mathbf{Z}$; it meets D_{3q} exactly in its two entries congruent to ± 1

(mod q). Hence λ_{3q} vanishes on S_{3q}. The first degree-33 and first degree-39 tuples in (3.1), $w_{33} = (1, 4, 16, 22, 25, 31)$ and $w_{39} = (1, 7, 16, 22, 34, 37)$, meet D_{3q} only in the entry 1, so $\lambda_{3q}(\nu(w_{3q})) = 1$. Thus the respective tuples generate the nontrivial integral gap classes, and

$$B_{3q} = S_{3q} + \mathbf{Z}\nu(w_{3q}). \tag{5.2}$$

Both w_{3q} are algebraic by Section 3. The detectors (5.1), their vanishing on S_{3q} and the resulting equality (5.2) are [5, Lemma 4.4 and Corollary 4.6], which are applied there to the same two tuples in the proof of [5, Theorem A]. Since D_{3q} is the subgroup of units with square one, for an odd squarefree semiprime degree m' the underlying criterion, that a Hodge element lies outside the subgroup $S_{m'}$ generated by inverse pairs and standard elements exactly when some coset of $\{\pm1\}\times\{\pm1\}$ contains an odd number of its entries, is [15, Theorem A(a)], which attributes it, in other terms, to Kubert. We include the argument only in the notation used here.

For $m = 3^a q^b$ with both exponents positive, [1, Theorem D, p. 38] generates B_m, as a U_m-module, by standards, pairs, and inflations of fixed $\xi_d \in B_d$, with $d \mid m$, $\mu'(d) = 1$. Unit translates preserve algebraicity by the algebraic orbit projectors described in Section 2. At odd level, the modified Möbius function is zero unless the divisor is squarefree, and on a squarefree divisor it is $\mu'(d) = (-1)^{\#\{p:p\mid d\}}$ [1, (3.3), p. 32]. Indeed, writing $d = \prod_{p\mid d} p^{e_p}$, the definition requires every $e_p = 1$ for a nonzero value; the convention is $\mu'(1) = 1$. The same odd-level identification with the usual Möbius function is stated in [3, p. 181]. The squarefree divisors of $3^a q^b$ are $1, 3, q, 3q$, so the only nontrivial one with $\mu'(d) = 1$ is $d = 3q$. The divisor one contributes no generator; equivalently, the same generation statement is printed with $d > 1$ in [3, Theorem 2.1, p. 181]. By (5.2), the fixed ξ_{3q}, whatever its particular construction, is an integral combination of standards, pairs and w_{3q}. Their inflations remain algebraic by Lemmas 2.1 and 2.2. Indeed, at odd degree Aoki's standard admissibility $pi \neq 0 \pmod m$ [1, p. 36] is equivalent to $d/\gcd(i, d) > 2$, since this odd quotient differs from one exactly when $d \nmid i$. For every inflation factor k, the multiset identity $k\sigma_{p,i}^{(m)} = \sigma_{p,ki}^{(km)}$ preserves that admissibility. Thus higher prime powers introduce no extra standard parameter hypothesis in this argument.

To pass from an integral identity to algebraicity, move negative coefficients to the other side. For a Hodge tuple T, this gives an equality of nonnegative multiplicity vectors $T \sqcup T_- = T_+$, where T_- and T_+ concatenate known algebraic Hodge tuples. Join the factors and cancel T_- by Section 2.3. This proves the assertion in every even dimension; non-middle classes in odd dimensions are ambient. If an exponent vanishes, [2, Corollary 2-3, p. 388] gives all dimensions for every prime-power degree, with no bound on its exponent. Its printed proof invokes Theorem 2-1, which assumes $\gcd(i, d) = 1$; the standard elements with $\gcd(i, d) > 1$ are covered by Lemma 2.2. Proposition 2.4 now proves products at one common degree. For different degrees, take their least common multiple M within this same fixed family, pull back to the product of degree-M varieties, and push forward divided by the product degree. □

The argument does not assert mixed products involving both an 11-supported and a 13-supported degree: their least common multiple can have three distinct prime divisors. No all-powers assertion for every odd degree follows from the fourfold classification alone.

6. A tensor filtration identity on a Fermat curve times a threefold

For a rational pure Hodge structure H, write $F^p_{\mathbf{Q}}H$ for its largest rational sub-Hodge structure contained in the complex Hodge filtration $F^pH_{\mathbf{C}}$. This is the rational level filtration of [9, pp. 519–520].

Proposition 6.1. For every odd $m \geq 5$, put $C = X^1_m$, $Y = X^3_m$, and $H = H^1(C) \otimes H^3_{\mathrm{prim}}(Y)$. Then

$$H^1(C) \otimes F^1_{\mathbf{Q}}H^3_{\mathrm{prim}}(Y) \subsetneq F^1_{\mathbf{Q}}H. \tag{6.1}$$

Proof. Let $g \in G_m$ act on the line of character υ by $g^* = \chi_\upsilon(g)$, and put $p(\upsilon) = \mathrm{age}_m(\upsilon) - 1$, the second Hodge index of the character line. Take

$$A = (1, 1, m-2), \qquad B = (2, 2, 2, m-5, m-1) \tag{6.2}$$

for $m \geq 7$. The diagonal coefficient-Galois orbit of the pair (A, B) is a rational sub-Hodge structure in H. If $1 \leq t < m/2$, then $p(tA) = 0$, and direct residue summation gives

$$p(tB) = 1 + \lfloor 5t/m \rfloor \in \{1, 2, 3\}. \tag{6.3}$$

There is no zero coordinate: $m \nmid 5t$ for a unit t when $m \geq 7$. Complex conjugation replaces the total second index by four minus that index. Thus all lines of this orbit have both Hodge indices at least one, and the entire rational orbit lies in $F^1_{\mathbf{Q}}H$.

On the other hand, $t = (m+1)/2$ is a unit and transforms B to

$$(1, 1, 1, (m-5)/2, (m-1)/2),$$

of age one. The orbit of B therefore contains an extreme $(3,0)$-line. It is stable under $t \mapsto -t$, so it also contains the conjugate $(0,3)$-line, which does not lie in F^1. Hence it is excluded from $F^1_{\mathbf{Q}} H^3_{\mathrm{prim}}(Y)$. The rational filtration is deck-stable; its complexification is a sum of character lines, so this exclusion also excludes the displayed tensor orbit from the left side of (6.1).

For $m = 5$, use $A = (1,1,3)$, $B = (2,2,2,2,2)$. As $t = 1,2,3,4$, the two second-index lists are $(0,0,1,1)$ and $(1,3,0,2)$; their sums are $(1,3,1,3)$. The same conclusion follows. □

The original homological notation in [9, proof of Proposition 2.2(iv), p. 522] gives the second equality

$$\bigoplus_{j+\ell=i} \mathcal{F}_h(W_{-j}H_j(X_1,\mathbf{Q}) \otimes W_{-\ell}H_\ell(X_2,\mathbf{Q})) = \bigoplus_{j+\ell=i} \bigoplus_{t+s=h} \mathcal{F}_t W_{-j}H_j(X_1,\mathbf{Q}) \otimes \mathcal{F}_s W_{-\ell}H_\ell(X_2,\mathbf{Q}). \tag{6.4}$$

We write h for the source's filtration index m, to distinguish it from the Fermat degree. The calligraphic $\mathcal{F}$ is the rational level filtration, as defined on pp. 519–521, not the complex Hodge filtration. For a smooth projective variety of dimension d, Poincaré duality gives $H_j = H^{2d-j}(d)$. Since [9, p. 521] defines $\mathcal{F}_m W_{-j}H_j = \mathcal{F}^{-m}W_{-j}H_j$ and a Tate twist (d) shifts the filtration index by d, this gives $\mathcal{F}_t H_j = F^{d-t}_{\mathbf{Q}} H^{2d-j}(d)$, in agreement with [9, (2.1), p. 521]. Apply (6.4) to the Künneth summand $(j,\ell) = (1,3)$ for $(X_1,X_2) = (C,Y)$, with $i = 4$, $h = 3$, and remove the common Tate twist (4). Since $F^1_{\mathbf{Q}}H^1(C) = 0$ and $F^2_{\mathbf{Q}}H^3(Y) = 0$, the only nonzero factor convolution is $H^1(C) \otimes F^1_{\mathbf{Q}}H^3(Y)$. Applying the primitive projector on Y would identify the two sides of (6.1). Proposition 6.1 disproves that equality on the actual Fermat product used in [10, equation (3.3), p. 227]. It does not assert a counterexample to the generalized Hodge conjecture.

The displayed character pair need not survive the particular invariant projection used to construct a fourfold from that product. The counterexample concerns the product identity and its proof; no additional assertion about its image under that projection is needed or made. A replacement argument for the product estimate could still establish conclusions for which this printed proof is insufficient.

7. Positive classification by conductor

The finite residual classification in Section 4 suggests the stronger statement that, at every odd degree, all sextuples outside Dec and the exact 5-standard family are inflations of (3.1). This is Theorem A, proved in Section 7.8 as Theorem 7.20. We first prove the subfamily results used in that reduction. For squarefree semiprimes we work in the integral character group and retain a length estimate. The argument below is a consequence of Aoki's character kernel; the resulting family is also covered by [11] and [5]. Lemmas 7.1 and 7.2 are not new. For an odd squarefree semiprime degree, the decomposition (7.2)–(7.3) is [15, Theorem 3.1], and (7.5) is the lower bound of [15, Theorem B]. The local term of that bound at each of the two primes equals the corresponding term $\delta_p(q)$ or $\delta_q(p)$ of (7.5), because -1 is a power of q modulo p exactly when $\mathrm{ord}_p(q)$ is even. [15] states the bound for every Hodge element outside the subgroup generated by inverse pairs and standard elements, without restriction to sextuples, and proves that it is attained, by constructions it attributes to [4, Lemmas 8.1 and 8.3]; [15] appeared on 22 October 2026, before the present version of these lemmas. We include short proofs in the notation used below; only the consequence for sextuples, $L(p,q) \le 6$ in (7.5), is used.

Put $m = pq$, with $p < q$ odd primes. For a tuple T, write n_x for its multiplicities and

$$F(x) = n_x - n_{-x}.$$

Thus F is an integer odd function, and

$$|T| \;\ge\; \frac{1}{2} \sum_{x \ne 0} |F(x)|. \tag{7.1}$$

The Hodge equations are equivalent to $\sum_x F(x) b_m(tx) = 0$ for every unit t, where $b_m(x) = [x]_m/m - 1/2$. An integer function F on $\mathbf{Z}/m\mathbf{Z} \setminus \{0\}$, or a multiset with multiplicity function F, satisfying these equations is called *balanced*; this term is used in this sense throughout Section 7. At $t = 1$ a balanced multiset T has $\sum_{x \in T}[x]_m = |T|m/2$, so a balanced multiset of even length has zero sum modulo m.

Lemma 7.1. In Chinese remainder coordinates on U_m, every integer-valued odd balanced function F, not necessarily coming from a multiset, has the unique form

$$F(a,b) = u(a) + v(b), \qquad a \in U_p,\ b \in U_q, \tag{7.2}$$

where u, v are rational odd functions. On the other two strata,

$$F(qy) = u(qy) - u(y), \qquad F(pz) = v(pz) - v(z). \tag{7.3}$$

Either both functions are integer-valued or both are valued in $\mathbf{Z} + \frac{1}{2}$. Define $\lambda(F) = 0$ or 1 accordingly. Then $\lambda(F) = 0$ if and only if F, modulo inverse pairs, is an integral combination of prime-standard tuples.

Proof. Let $\chi = (\chi_p, \chi_q)$, with both local components nonprincipal. It is primitive of conductor exactly m. For odd χ, its Fourier coefficient is

$$\sum_{x \in U_m} \chi(x) b_m(x) = \frac{1}{m} \sum_{x \in U_m} \chi(x)[x]_m = S(\chi) \neq 0$$

by [1, Lemma 2.3, p. 29]. In the multiplicative Fourier transform of the balance equations the order-p and order-q strata have support only on $(\chi_p, 1)$ and $(1, \chi_q)$, respectively. Thus the unit-stratum correlation has coefficient $(\sum_x F(x)\bar{\chi}(x))S(\chi) = 0$. For even χ, oddness of F already makes its coefficient zero. Hence the Fourier coefficients of the unit matrix vanish whenever both local characters are nonprincipal. The remaining Fourier modes are exactly the sum of a row function and a column function. Taking averages in each coordinate gives unique zero-mean functions u, v; oddness of F makes each odd.

For fixed $a \in U_p$, the distribution identity is

$$\sum_{b \in U_q} b_m(\mathrm{CRT}(a,b)) = b_p(a) - b_p(q^{-1}a). \tag{7.4}$$

Indeed, the sum over all q lifts of a is $b_p(a)$, and the omitted lift divisible by q contributes $b_p(q^{-1}a)$. Substituting (7.2) into the balance equations, (7.4) gives (7.3): the remainder is a sum of a function of the p-unit and a function of the q-unit. More explicitly, put $D(y) = F(qy) + u(y) - u(qy)$. Its correlation $f_p(t) = \sum_y D(y) b_p(ty)$ is odd and independent of the q-unit. Since $f_p(t_p) + f_q(t_q) = 0$ for independently varying units, both functions are constant, hence zero by oddness. Every odd character modulo the prime p is primitive; the same nonvanishing therefore gives $D = 0$. The argument on the other stratum is identical. This proves (7.3).

The equalities $F(a,b) + F(a,-b) = 2u(a)$ and $F(a,b) - F(a',b) = u(a) - u(a')$ show that $2u$ is integral and all values of u have the same fractional part. Oddness makes that part zero or one half. The same holds for v, and the two parts agree because $F(a,b)$ is integral.

The unit part of a q-standard tuple with parameter congruent to $a_0 \pmod p$ has $u = \mathbf{1}_{a_0} - \mathbf{1}_{-a_0}$, $v = 0$. These are admissible standards since $p > 2$ and the parameter is nonzero modulo p. Likewise p-standards generate all integer odd column functions. Equations (7.3) then determine the lower strata. Thus integer-valued u, v are exactly the integral span of standards in the quotient by inverse pairs, proving the final assertion. □

Here equality modulo inverse pairs is equality in the free abelian group on nonzero residues modulo $(x) + (-x) = 0$. Thus an integral relation may have signed coefficients. Moving its negative pair coefficients across the equality yields an equality of nonnegative multisets after replacing negative standards by their conjugates. Oddness of m ensures that no nonzero residue is its own negative.

For a prime r and an integer s prime to it, set

$$\delta_r(s) = \begin{cases} (r-1)/\operatorname{ord}_r(s), & \operatorname{ord}_r(s) \text{ even}, \\ 0, & \operatorname{ord}_r(s) \text{ odd}. \end{cases}$$

Lemma 7.2. If $\lambda(F) = 1$, then

$$|T| \geq L(p,q) := \frac{(p-1)(q-1)}{4} + \delta_p(q) + \delta_q(p). \tag{7.5}$$

Proof. For each pair of antipodal pairs $\{a,-a\}$, $\{b,-b\}$, the contribution of their four unit entries to the right side of (7.1) is

$$|u(a)+v(b)|+|u(a)-v(b)| = 2\max(|u(a)|,|v(b)|) \geq 1.$$

There are $(p-1)(q-1)/4$ such blocks. On the order-p stratum, (7.3) contributes $\frac{1}{2}\sum_y |u(qy)-u(y)|$. If the order of q modulo p is even, each multiplication-by-q cycle contains y and $-y$. The half-integral odd values include opposite nonzero values, so total variation around each cycle is at least two. There are $(p-1)/\operatorname{ord}_p(q)$ cycles, giving $\delta_p(q)$. If that order is odd the asserted lower bound on this stratum is zero. The order-q stratum gives the remaining term. □

Proof of Theorem 1.2. If $p \geq 5$ and $(p,q) \neq (5,7)$, the first term of (7.5) already exceeds six. For $(5,7)$, both orders are even and $L(5,7) = 8$. If $p = 3$, then the first term exceeds six as soon as $q \geq 17$. Thus a Hodge sextuple with $\lambda = 1$ is possible only at degrees $15, 21, 33, 39$.

If $\lambda = 0$, Lemma 7.1 expresses the tuple as an integral combination of algebraic standard tuples modulo inverse pairs. Such an identity implies algebraicity: replace any negative standard by its coordinatewise negative, which is a Galois conjugate and the negative of its class modulo pairs. The resulting nonnegative concatenation and the given tuple differ by a signed sum of inverse pairs. Put the negative pairs on the other side, join the known algebraic factors, and cancel the remaining known pair factors as in Section 2.3.

At degrees 15 and 21 the surface Hodge tuples $(1,7,10,12)$ and $(1,10,15,16)$ have $\lambda = 1$. These reference tuples have length four, whereas the tuple being treated has length six; the joins and cancellations take place in a larger auxiliary dimension, so their lengths need not agree. At 33 and 39 use the first tuples in (3.1), algebraic by Section 3. In the coordinates of (7.2), the points $(1,1)$ and $(1,-1)$ are the residues 1 and 4, 1 and 13, 1 and 10, 1 and 25 at degrees 15, 21, 33, 39 respectively. Each reference tuple contains the first of these residues and not the second, nor the negative of either, so $\lambda = 1$ for it. The second degree-39 seed has $\lambda = 0$ and is not used. These parity assertions follow either from (7.2) or from the formula

$$\lambda(F) = F(1,1) + F(1,-1) \pmod 2.$$

Subtracting the corresponding known algebraic tuple from a $\lambda = 1$ sextuple gives an integer-potential balanced function, which is in the standard span by Lemma 7.1. Apply the same integral realization argument, now allowing the one additional known tuple. This proves the theorem. □

The estimate and its normalization have independent finite checks: all half-sign potentials for $(3,5),(3,7),(3,11),(3,13),(3,17),(5,7)$ give integral balanced functions with the predicted stratum equations; the minimum lengths are respectively $4,4,6,6,10,8$. These are checks of the formulas, not the proof of their unrestricted range.

7.1. Positive classification in semiprime degree

Integral generation alone does not imply a positive decomposition of a sextuple. The length-six constraint supplies the additional argument in semiprime degree.

Theorem 7.3. If $m = pq$, with $p < q$ odd primes, every Hodge sextuple outside $\mathrm{Dec} \cup \mathrm{Std}_5$, modulo units and coordinate permutations, is the degree-33 tuple or one of the two degree-39 tuples in (3.1). In particular, no new primitive residual occurs at any squarefree odd semiprime degree.

Proof. A tuple containing an inverse pair is already in Dec. Assume it is pair-free. Then F determines the tuple uniquely as its positive part, and (7.1) is an equality. Write $A = (p-1)/2$, $B = (q-1)/2$, and choose representatives a_i, b_j of the antipodal pairs in the two prime unit groups. The unit stratum contributes exactly

$$N_U = 2\sum_{i=1}^{A}\sum_{j=1}^{B} \max(|u(a_i)|,|v(b_j)|) \tag{7.6}$$

to the length. For $p \geq 7$, [1, Theorem A] already makes every Hodge sextuple a union of inverse pairs.

Suppose $p = 5$. Lemma 7.2 gives $\lambda = 0$: its bound is eight at $q = 7$ and exceeds six for $q \geq 11$. If $q \geq 11$, a nonzero integer u would give $N_U \geq q-1 > 6$, so $u = 0$. At $q = 7$, a nonzero u forces $v = 0$ and precisely one nonzero antipodal value of u, equal to ± 1; otherwise (7.6) is at least eight. But multiplication by $7 = 2 \pmod 5$ moves its antipodal support, so the order-5 stratum contributes two more entries, again impossible. Thus $u = 0$ in this case as well. Equation (7.6) now forces $v = e_b$ for one nonzero b, where $e_b = \mathbf{1}_b - \mathbf{1}_{-b}$. This is exactly the odd multiplicity

function of a 5-standard tuple. Its two lower-stratum entries do not cancel, because $5 \not\equiv \pm 1 \pmod q$. Consequently that standard tuple is pair-free of length six, and uniqueness of the positive part identifies it with the original tuple.

Suppose $p = 3$, $q \geq 17$. Lemma 7.2 again gives $\lambda = 0$, and (7.6) gives $u = 0$. Set

$$r = \sum_{j=1}^{B} |v(b_j)|, \qquad d(v) = \frac{1}{2} \sum_{z \in U_q} |v(3z) - v(z)|.$$

Then $6 = 2r + d(v)$, so $1 \leq r \leq 3$. If $r = 1$, then $v = e_b$ and $d(v) = 2$, giving length four rather than six. If $r = 3$, then $d(v) = 0$. The positive support of a nonzero invariant v is a union of multiplication-by-3 cycles of total multiplicity at most three. Thus $\operatorname{ord}_q(3) \leq 3$, which would force $q \mid 3^j - 1$ for $j \leq 3$; no prime $q \geq 17$ does so.

It remains that $r = 2$, $d(v) = 2$. A value of absolute size two at a single antipodal pair would instead give $d(v) = 4$. Hence $v = e_b + e_c$, $b \neq \pm c$. The four signed atoms $e_{b/3}, e_{c/3}, -e_b, -e_c$ in $v(3z) - v(z)$ have half-norm less than four only if two of them cancel. The cancellations $b/3 = b$ and $c/3 = c$ force $3 \equiv 1 \pmod q$, and $b/3 = -c/3$ or $b = -c$ contradict $b \neq \pm c$. The only remaining cancellations are $b/3 = c$ and $c/3 = b$, that is $b = 3c$ or $c = 3b$ (both together would force $9 \equiv 1$). Relabel so that $v = e_b + e_{3b}$. Choose a 3-standard tuple with parameter congruent to $b \pmod q$, and another with parameter congruent to $3b \pmod q$. Each is a surface Hodge quadruple. The last coordinate $-3i$ of the first cancels the unique coordinate divisible by 3 in the progression of the second: both have opposite residues modulo q and vanish modulo 3. Removing this inverse pair leaves six entries with the required odd function. Their norm is six, so they are pair-free and equal to the original tuple. This is precisely a surface contraction, hence belongs to Dec.

The remaining cases are $p = 3$, $q \in \{5, 7, 11, 13\}$, that is, the degrees $15, 21, 33, 39$. Here $A = 1$; put $\mu = u(1) = -u(-1)$. By (7.3) the order-3 stratum contributes $N_3 = 0$ if $q \equiv 1 \pmod 3$ and $N_3 = 2|\mu|$ if $q \equiv 2 \pmod 3$. With $d(v)$ as above, now for possibly half-integral v,

$$6 = N_U + N_3 + d(v), \qquad N_U = 2 \sum_{j=1}^{B} \max(|\mu|, |v(b_j)|). \tag{7.6a}$$

Throughout, 3 generates U_5 and U_7, has order five modulo 11 and order three modulo 13, and $3 \not\equiv \pm 1$, $9 \not\equiv 1 \pmod q$.

Case $\lambda = 0$, $\mu = 0$. Then $N_3 = 0$, $N_U = 2r$, and the argument given for $q \geq 17$ applies without change: $r = 1$ gives length four, $r = 2$ gives a surface contraction, and $r = 3$ requires $q \mid 3^j - 1$ with $j \leq 3$, so $q = 13$. At $q = 13$ the positive support of v is one of the four cosets of $\langle 3 \rangle = \{1, 3, 9\}$; the unit congruent to 1 modulo 3 and to 2 modulo 13 permutes them transitively, since 2 generates $U_{13}/\langle 3 \rangle$. Thus $v = e_1 + e_3 + e_9$ up to units, $F(\pm 1, b) = v(b)$, and the tuple is the second degree-39 tuple $(1, 14, 16, 22, 29, 35)$ of (3.1).

Case $\lambda = 0$, $\mu \neq 0$. Now $N_U \geq (q-1)|\mu|$, so $q \in \{5, 7\}$, $|\mu| = 1$, and after multiplication by -1, $\mu = 1$. At $q = 5$, $N_U \geq 4$ and $N_3 = 2$; at $q = 7$, $N_U \geq 6$ and $N_3 = 0$. In both cases $d(v) = 0$, so v is invariant under multiplication by a generator of U_q, hence constant, hence zero by oddness. The unit entries are the $q - 1$ units congruent to 1 modulo 3, and (7.3) supplies the order-3 entries. At 15 this is $(1, 4, 7, 10, 10, 13) = \sigma_{5,1}$. At 21 it is $(1, 4, 10, 13, 16, 19)$, the contraction of the surface tuples $(1, 10, 16, 15)$ and $(6, 4, 13, 19) = 13 \cdot (1, 10, 15, 16)$ along the distinguished residues $15, 6$, with $1 + 10 + 16 \equiv 6 \pmod{21}$.

Case $\lambda = 1$. Since $N_U \geq (q-1)|\mu|$, the value $|\mu| \geq \frac{3}{2}$ would give $N_U + N_3 \geq 9 > 6$. Hence $|\mu| = \frac{1}{2}$, and after multiplication by -1, $\mu = \frac{1}{2}$. Every $|v(b)| \geq \frac{1}{2}$, so $N_U \geq (q-1)/2$, with equality exactly when $|v| = \frac{1}{2}$ everywhere. At $q = 13$, (7.6a) forces equality, $N_3 = 0$ and $d(v) = 0$. Thus $v = \pm\frac{1}{2}$ is constant on the cosets of $\langle 3 \rangle$. The four sign patterns form one orbit under the unit used above, which sends the values $(\varepsilon_1, \varepsilon_2)$ on $\{1, 3, 9\}$ and $\{2, 5, 6\}$ to $(-\varepsilon_2, \varepsilon_1)$. For $v = \frac{1}{2}$ on $\{1, 3, 9\} \cup \{7, 8, 11\}$, the function $F(1, b) = \frac{1}{2} + v(b)$ is the indicator of this set and the lower strata vanish, giving $(1, 7, 16, 22, 34, 37)$. At $q = 11$, $N_3 = 1$, so $N_U = 5$ and $d(v) = 0$. Now $\langle 3 \rangle$ is the group of squares, of index two, and its two sign patterns are exchanged by the unit congruent to 1 modulo 3 and to 2 modulo 11. For $v = \frac{1}{2}$ on the squares the unit entries are the five units congruent to 1 modulo 3 whose residues modulo 11 are squares, and (7.3) gives $F(11) = u(-1) - u(1) = -1$, that is, the entry 22. This is $(1, 4, 16, 22, 25, 31)$.

At $q = 5, 7$, let $a_k = v(3^k)$, so that $a_{k+(q-1)/2} = -a_k$ and $d(v) = \sum_{k=0}^{(q-3)/2} |a_{k+1} - a_k|$. The triangle inequality, applied from each starting index, gives $d(v) \geq 2 \max_k |a_k|$, and $N_U = 2 \sum_j |v(b_j)|$. At $q = 5$ this makes the length

$1+2(|a_0|+|a_1|)+2\max(|a_0|,|a_1|)$, which is four when $|a_0| = |a_1| = \frac{1}{2}$ and at least eight otherwise; there is no sextuple. At $q = 7$, $N_3 = 0$, and a value of absolute size at least $\frac{3}{2}$ would give length at least eight. Hence $|a_0| = |a_1| = |a_2| = \frac{1}{2}$ and $d(v) = 3$, which forces $a_1 = -a_0$, $a_2 = a_0$. Multiplication by the unit congruent to 1 modulo 3 and to 3 modulo 7 exchanges the two choices of a_0. For $a_0 = \frac{1}{2}$ the tuple is $(1, 4, 9, 15, 16, 18)$, the contraction of $(1, 15, 16, 10)$ and $\sigma_{3,4} = (11, 4, 9, 18)$ along the distinguished residues $10, 11$. It lies in Dec.

Hence the pair-free Hodge sextuples outside $\mathrm{Dec} \cup \mathrm{Std}_5$ at these four degrees lie in the three stated orbits. They are outside Dec: at $m = 3q$, $q \in \{11, 13\}$, apply the same analysis to a pair-free balanced quadruple. Lemma 7.2 gives $L(3, q) = 6 > 4$, so $\lambda = 0$; $N_U \geq q - 1$ forces $\mu = 0$; and $4 = 2r + d(v)$ leaves $r = 1$, because $r = 2$, $d(v) = 0$ would need $q \mid 3^j - 1$ with $j \leq 2$. Hence every pair-free surface Hodge quadruple at 33 or 39 has exactly two unit entries. A balanced quadruple containing an inverse pair is a union of two inverse pairs, so a pair-free sextuple in Dec is a contraction of two pair-free quadruples and has at most four unit entries. The three tuples have five, six and six unit entries. They are not 5-standard, as $5 \nmid m$. The two degree-39 orbits are distinct, because the units preserve λ, which is 1 for the first and 0 for the second. □

7.2. Positive classification at prime powers

Proposition 7.4. Let $m = p^a$ be an odd prime power. Every Hodge sextuple lies in $\mathrm{Dec} \cup \mathrm{Std}_5$.

Proof. At a prime degree p every nonzero residue is a unit and every odd character is primitive, so [1, Lemma 2.3] and the balance equations make all odd Fourier coefficients of the odd multiplicity function of a pair-free tuple vanish; that function is then zero, which is impossible. Thus every Hodge sextuple at prime degree contains an inverse pair; this is also the case $n = 4$ of [1, Theorem A(i)]. Reduce content and discard the case of an inverse pair. It suffices to consider a primitive pair-free tuple at $a \geq 2$. At a primitive character modulo p^a, every lower-order stratum vanishes from the Fourier transform of the balance equations. Primitive odd nonvanishing [1, Lemma 2.3] and oddness therefore force the unit-stratum function F to have Fourier support of conductor at most p^{a-1}. It is constant on each fibre of $U_{p^a} \to U_{p^{a-1}}$, of size p. Pair-freeness implies the same constancy for positive multiplicities. Thus the unit entries are complete progressions

$$P_i = (i, i + d, \ldots, i + (p-1)d), \qquad d = p^{a-1}, \quad p \nmid i,$$

counted with multiplicity. At least one occurs by primitiveness. For every unit t, the distribution identity is

$$\sum_{x \in P_i} b_m(tx) = b_m(tpi). \tag{7.7}$$

Replacing a progression by pi preserves balance and decreases length by $p - 1$; all resulting entries are nonzero and divisible by p.

If $p \geq 7$, a progression is already too long. If $p = 5$, there is exactly one progression and one other entry. Replacement gives a balanced pair, necessarily inverse, so the original sextuple is exactly $\sigma_{5,i}$. If $p = 3$ and there are two progressions, replacement gives a balanced pair $(3i, 3j)$. Therefore $i + j = 0 \pmod{d}$, making the two progressions inverse to each other, contrary to pair-freeness (a repeated progression would require $2i = 0 \pmod{d}$). If there is just one progression, the other three entries, together with $3i$, form a surface Hodge quadruple by (7.7). Its contraction with the standard surface tuple $(P_i, -3i)$ recovers the sextuple, proving Dec. Content inflation preserves both criteria, as proved in Section 4. □

Theorem 7.3 and Proposition 7.4 establish the positive six-seed classification on these two infinite conductor classes. Sections 7.3–7.4 extend it to further conductor classes, before Section 7.8 proves the unrestricted six-seed statement.

7.3. Degrees coprime to six

The prime-to-six branch is also treated in [5, subsection "Fermat fourfolds of degree prime to six"], whose finite-product argument is the same in spirit; the case is also covered by [11]. We give a self-contained proof of the exact positive classification used here; neither all-odd theorem is a premise.

Call a function on a finite product $C = C_1 \times \cdots \times C_r$ split if it is a sum of functions, each independent of at least one coordinate. No integrality of the summands is assumed.

Lemma 7.5. Suppose $|C_i| \geq 4$, with at most one factor of size four. An integer-valued, nonzero split function f with $\sum_C |f| \leq 6$ is 1 or -1 on one coordinate line and zero elsewhere.

Proof. Induct on the number of factors. There is no nonzero function for zero factors; on one factor split means constant. Slice along the first factor. Differences of two slices are split on the remaining factors. If one slice vanishes, every slice is split on the remaining factors. By induction, each nonzero slice has mass at least four and is a signed line. At most one such slice can occur, proving the assertion. If all slices agree, invariance in the first coordinate and the mass bound give one line with value ± 1.

It remains that every slice is nonzero and they do not all agree. Some slice has mass one, since $6 < 2|C_1|$. If all slices had mass one, their pairwise differences would have mass at most two; induction would make every difference zero. Therefore a slice of mass $M \geq 2$ exists. Its difference from a mass-one slice is nonzero and split, so induction gives $M + 1 \geq |C_j| \geq 4$ for some $j > 1$. Hence $6 \geq M + |C_1| - 1 \geq 3 + 3$. Equality forces both $|C_1| = 4$ and $|C_j| = 4$, contrary to the hypothesis. □

Theorem 7.6 (prime-to-six classification). If $\gcd(m, 6) = 1$, every Hodge sextuple contains an inverse pair or is exactly 5-standard. In particular the rational Hodge conjecture holds for X_m^4, and $\mathcal{R}(m) = \varnothing$ for the residual set in Section 2.3.

Proof. Let T be pair-free, let e be its largest additive order, and identify the subgroup of order e in $\mathbf{Z}/m$ by $\kappa(z) = (m/e)z$. On U_e put $F(z) = n_{\kappa(z)} - n_{-\kappa(z)}$. Then F is odd, nonzero, and has mass at most twelve. A primitive odd character of conductor e annihilates every other order stratum in the Fourier transform of balance: an order smaller than e cannot be a multiple of e. Nonvanishing [1, Lemma 2.3] therefore makes every primitive Fourier coefficient of F zero; even coefficients vanish by oddness.

For each prime $p \mid e$, let $K_p = \ker(U_e \to U_{e/p})$. These groups have direct product H, and sizes p if $p^2 \mid e$, $p-1$ otherwise. A nonprimitive character is trivial on some K_p, so Fourier inversion makes F split on every coset of H. Every size is at least four, and only K_5 can have size four. If e is prime, split means constant on U_e, contradicting oddness and nonvanishing. Thus e is composite.

We first show that $F \geq 1$ on a line wK_p with $p \in \{5, 7\}$, apart from a possible full-mass case at $e = 35$. If $-1 \notin H$, each coset and its negative have the same mass, at most six each. Lemma 7.5 gives a signed line; negating the coset if needed makes its sign positive. Its size is at most six, forcing $p = 5$ or 7. If $-1 \in H$, then e is squarefree: a repeated prime would make all of H congruent to one at that prime. Thus $H = U_e$.

Suppose a prime $p \geq 11$ divides this squarefree e. There is at least one other prime, since a split odd function on a single factor is zero. Slice along K_p, of size at least ten. Oddness pairs the slices, preserving their masses. If a slice is zero, restriction to any other slice is split in the remaining factors and has mass at most six; Lemma 7.5 again gives the desired line. If every slice is nonzero, the total mass (at most twelve, and even) and pairing force every slice mass to be at most two, with some mass one. The difference between that slice and any other has mass at most three, so Lemma 7.5 forces it to vanish. Hence all slices have one equal signed atom: the support is one K_p-line, with constant sign. Its fixed coordinate in the remaining factors differs from its negative, because those factors have odd residue characteristic. Oddness would require another supported line there, a contradiction.

The remaining squarefree composite is $e = 35$. Its split odd matrix has the form $u(a) + v(b)$ on $U_5 \times U_7$, with odd potentials simultaneously integral or half-integral, by the row-and-column argument of Lemma 7.1. In the integral case, (7.6) shows that both potentials cannot be nonzero: that would cost at least eight entries. A single nonzero potential must be one signed antipodal atom, giving the required positive line. In the half-integral case the unit contribution is at least six, hence the full mass is twelve and all six entries of T have order 35. They divide via κ to a pair-free all-unit Hodge sextuple at degree 35. The proof of Theorem 7.3 at $(p, q) = (5, 7)$ excludes this: every pair-free sextuple there is 5-standard, and a 5-standard sextuple at 35 has two nonzero entries divisible by 5, so it is not all-unit. Thus this case is impossible.

Consider the positive line obtained above. For $p = 7$ its size cannot be seven, and size six consumes the whole mass twelve. Thus all six entries have order e, F is 1 on the line and -1 on its negative. They are disjoint, since oddness forbids $-1 \in K_p$. The zero sum of the tuple gives $\sum_z F(z)z = 0 \pmod{e}$, whereas reduction modulo $e/7$ gives $12w$, since every $k \in K_7$ is one modulo $e/7$. Here $e/7 > 1$ is coprime to six, so cannot divide twelve. Therefore $p = 5$.

Put $x = \kappa(w)$, and $P_x = (x, x + m/5, \ldots, x + 4m/5)$. The line supplies either five or four entries of this progression. In

the first case the sixth entry is $-5x$, so the tuple is exactly standard. In the second, write $T \sqcup \{y\} = P_x \sqcup \{b, c\}$. We have $5x \neq 0$, because $e \neq 5$; and $y \neq 5x$, since otherwise $20x = 0$ would force $5x = 0$ at odd degree. The distribution identity for P_x shows that $Q = (5x, b, c, -y)$ is a nonzero surface Hodge quadruple. Every prime divisor of m exceeds four, so [1, Theorem A, with hypersurface dimension $n = 2$] makes Q a union of inverse pairs. Its pair cannot be $(5x, -y)$, or (b, c), by the two exclusions just made and the pair-freeness of T. Any other pair, together with the zero sum, gives $\{b, c\} = \{y, -5x\}$. Thus $T = P_x \sqcup \{-5x\}$ in every case. Since $5x \neq 0$, the tuple is admissible by the remark after (2.4), and Lemma 2.2 gives its algebraicity. □

The last theorem and Proposition 7.4 classify conductors prime to six and pure prime-power depths. The next subsection treats two-prime depths containing 3, except for support $\{3, 5\}$. Bhattacharjee [5, subsection "Fermat fourfolds of odd degree divisible by three"] states a stronger profile descent whose terminal orbits are at 21, 33 and 39. We do not use that unreviewed all-odd theorem as a premise. Its profile descent also differs from the exact residual classification proved here. For example, the second degree-39 seed contains $1, 14$ of the 3-progression $\{1, 14, 27\}$. It admits a profile-lowering move in the sense of [5], although it lies outside $\mathrm{Dec} \cup \mathrm{Std}_5$.

7.4. Two-prime depths with a prime at least seven

We make the conductor calculation explicit. This is the character relation of [5, Lemma 3.28], proved there from $B_{1,\chi} \neq 0$; we derive it from the classical nonvanishing input [1, Lemma 2.3] and include its proof and normalization.

Lemma (conductor Fourier relation). Let T be a balanced multiset of nonzero residues modulo odd m. Write an entry as $a = (m/f_a)z_a$, where f_a is its additive order and $z_a \in U_{f_a}$. For an odd primitive Dirichlet character χ of conductor $c \mid m$,

$$\sum_{a \in T,\ c \mid f_a} \frac{\overline{\chi(z_a)}}{\varphi(f_a)} \prod_{\substack{p \mid f_a \\ p \nmid c}} (1 - \chi(p)) = 0. \tag{7.7a}$$

Primes, rather than prime powers, index the product.

Proof. Put $B_{1,\chi} = c^{-1} \sum_{r=1}^{c} r\chi(r)$. This is nonzero for a primitive odd character by [1, Lemma 2.3]. For every multiple f of c, periodicity and $\sum_{r=1}^{c} \chi(r) = 0$ give $\sum_{r=1}^{f} r\chi(r) = fB_{1,\chi}$. Excluding the primes dividing f but not c by inclusion and exclusion therefore gives

$$\sum_{r \in U_f} r\chi(r) = fB_{1,\chi} \prod_{p \mid f,\ p \nmid c} (1 - \chi(p)).$$

Here r denotes its representative in $\{1, \dots, f - 1\}$. If $c \nmid f_a$, the character χ is nontrivial on the kernel of $U_m \to U_{f_a}$, so averaging $\chi(t)[ta]_m$ over that kernel gives zero. Indeed, choose a prime with $v_p(f_a) < v_p(c)$. Primitivity makes the local character nontrivial on the kernel of reduction to this smaller prime-power modulus. Lift such a unit to U_m, taking its other prime components to be one; the lift is one modulo f_a and has nontrivial character value. Otherwise unit reduction is surjective and changing variables by z_a gives

$$\sum_{t \in U_m} \chi(t)[ta]_m = \frac{m\varphi(m)B_{1,\chi}}{\varphi(f_a)} \overline{\chi(z_a)} \prod_{p \mid f_a,\ p \nmid c} (1 - \chi(p)).$$

The balance equation makes the sum over a of the left side zero. Divide by the common nonzero factor to obtain (7.7a). □

An entry order is called maximal if no entry has order a proper multiple of it. The entries of a maximal order e are called the *tops* at e. At such an order e, equation (7.7a) with $c = e$ involves only that stratum. Let $F_e(z) = n_{(m/e)z} - n_{-(m/e)z}$. Its odd primitive Fourier coefficients vanish; its even coefficients vanish by oddness. For $K_p = \ker(U_e \to U_{e/p})$, a character is nonprimitive exactly when it is trivial on at least one K_p. Thus Fourier inversion writes F_e as a sum of functions invariant under one of the K_p. Restricting to a coset of their direct product H makes it split in the precise sense of Section 7.3. This proves the maximal-order assertion for arbitrary prime support and exponents, without assuming nonvanishing for imprimitive characters.

We will also use the following elementary correspondence between these lines and positive progressions. Every $k \in K_p$ is one modulo e/p, so $(m/e)wk - (m/e)w$ is a multiple of m/p. Consequently a positive wK_p-line gives p distinct points

of a p-progression if $p^2 \mid e$, and $p-1$ if $p \parallel e$. In the latter case the omitted point is the unique one whose order loses the p-factor. This is the fibre-line observation in [5, Lemma 3.29].

Lemma 7.7. Let C have $s \geq 6$ elements. An integer split function on $\{1,2,3\} \times C$, of mass at most six, is constant in the first coordinate if $s > 6$. If $s = 6$, either it is constant in that coordinate or it equals one signed atom in the first coordinate, repeated over all of C.

Proof. Write its entries as $f(i,c) = u_i + v_c$. If two rows differ, their difference is a nonzero integer, so each column has mass at least one. This is impossible for $s > 6$. For $s = 6$, all columns must have mass one. Two such integer column vectors differ by a constant vector. A nonzero constant difference would have mass at least three, whereas the difference of two mass-one vectors has mass at most two. Consequently all columns are equal to the same signed atom. □

Theorem 7.8. A primitive Hodge sextuple at degree $m = 3^a q^b$, with $q \geq 7$ prime and $a, b \geq 1$, $\max(a,b) > 1$, belongs to Dec.

Proof. Assume the tuple T is pair-free. First let $a \geq 2$. Choose a maximal order $e = 3^a q^c$ among entries with full 3-primary order; primitiveness guarantees such an entry. On its order stratum let F be the nonzero odd multiplicity function. Its mass is at most twelve. If $c = 0$, primitive Fourier vanishing makes F constant on the three-element fibres $U_e \to U_{e/3}$. A positive fibre is a complete 3-progression in T.

If $c > 0$, put

$$K_3 = \ker(U_e \to U_{e/3}), \qquad K_q = \ker(U_e \to U_{e/q}), \qquad H = K_3 K_q.$$

Their orders are three and s, where $s = q-1$ for $c = 1$ and $s = q$ otherwise. Fourier inversion makes F split on every coset of $H = K_3 \times K_q$. Since $9 \mid e$, $-1 \notin H$; a coset and its negative therefore each have mass at most six. Lemma 7.7 gives a positive complete K_3-line unless $q = 7, c = 1$ and the entire positive support is a single K_7-line of length six. In that exceptional case all entries have order e. Write them as $(m/e)wk$, $k \in K_7$. The zero sum of T, reduced modulo $e/7 = 3^a$, would give $6w = 0$, impossible since $a \geq 2$ and w is a unit.

Thus T contains a full progression $P = (x, x + m/3, x + 2m/3)$, with $3x \neq 0$. Replacing P by $3x$ gives a surface Hodge quadruple by the distribution identity (7.7). Its contraction with $(P, -3x)$ recovers T, proving $T \in \mathrm{Dec}$.

It remains that $a = 1, b \geq 2$. Put $Q = q^b$, and on the unit stratum write $F(s,u)$, with $s \in U_3$, $u \in U_Q$. Set $D(u) = F(1,u) - F(-1,u)$. This is even. A primitive character at degree $3Q$ is the product of the nontrivial character modulo 3 and a primitive character modulo Q. The odd such products use the even primitive characters of Q. Their vanishing says that every primitive Fourier coefficient of D is zero; its odd coefficients already vanish by evenness. Hence D is constant on fibres of $U_Q \to U_{Q/q}$, of size q. Each fibre is disjoint from its negative. A nonzero even function constant on these fibres has mass at least $2q > 12$, whereas $\|D\|_1 \leq \|F\|_1 \leq 12$. Thus

$$F(1,u) = F(-1,u) = v(u), \qquad v(-u) = -v(u), \quad v(u) \in \mathbf{Z}. \tag{7.8}$$

This conclusion also holds if the unit stratum was initially empty.

Let $G(y) = n_{3y} - n_{-3y}$ on U_Q, describing the order-Q stratum. Primitive odd Fourier coefficients at conductor Q give

$$G(y) = v(3y) - v(y). \tag{7.9}$$

Here is why this is an equality, including the possible imprimitive Fourier modes. The distribution identity

$$\sum_{s \in U_3} b_{3Q}(\mathrm{CRT}(s,u)) = b_Q(u) - b_Q(3^{-1}u) \tag{7.10}$$

shows that $E = G + v - v(3\,\cdot)$ has zero primitive odd Fourier coefficients. All other order strata have smaller q-primary order and contribute zero to those coefficients. Oddness removes the even coefficients, so E is fibre-constant as above. If N_U, N_Q are the numbers of entries in the two strata, then

$$\|E\|_1 \leq \|G\|_1 + 2\|v\|_1 = 2N_Q + 2N_U \leq 12.$$

A nonzero odd fibre-constant function also has mass at least $2q$. Thus $E = 0$, proving (7.9).

Equations (7.8)–(7.10) define by their positive parts a balanced submultiset $B \subseteq T$, entirely in these two strata. Indeed the correlations of the two odd functions cancel identically in (7.10). If $v = 0$, then $G = 0$ as well, and every entry has order dividing $3q^{b-1}$, contrary to primitiveness. Put

$$r = \tfrac{1}{2}\|v\|_1 \geq 1, \qquad d = \tfrac{1}{2}\sum_{y \in U_Q} |v(3y) - v(y)|.$$

Then $|B| = 2r + d \leq 6$, so $r \leq 3$.

For $r = 1$, write $v = e_z$, where $e_z = \mathbf{1}_z - \mathbf{1}_{-z}$. Since $3 \not\equiv \pm 1 \pmod{Q}$, $d = 2$ and $|B| = 4$. The remaining two entries would form a balanced pair, contradicting pair-freeness of T. For $r = 3$, the length bound forces $d = 0$. Its positive support would be a union of multiplication-by-3 cycles of total multiplicity three. Consequently $Q \mid 3^j - 1$ for some $j \leq 3$, impossible since $Q \geq 49$.

For $r = 2$, a repeated atom $v = 2e_z$ gives $d = 4$, which is too long. Otherwise $v = e_z + e_w$, $w \neq \pm z$. An odd function of norm four cannot be invariant under multiplication by 3 at $Q \geq 49$; hence $d > 0$. Cancellation in

$$e_{3^{-1}z} + e_{3^{-1}w} - e_z - e_w$$

is possible only if $w = 3z$ or $z = 3w$: the self-cancellation possibilities would require $3 \equiv \pm 1 \pmod{Q}$. With no cancellation its half-norm is four, exceeding the bound $d \leq 2$. Relabel so that $w = 3z$. The two 3-standard surface tuples whose parameters are congruent to z and $3z$ modulo Q have one distinguished inverse pair: the last coordinate of the first and the progression coordinate divisible by 3 of the second. Contracting that pair gives a sextuple with exactly (7.8)–(7.9). Its length is six, so it is pair-free and equals $B = T$. This is a surface contraction, as required. □

Proof of Theorem 1.3. Divide out the content of a residual sextuple. Proposition 7.4 handles a remaining prime power, Theorem 7.8 excludes primitive residuals at positive depths with $\max(a, b) > 1$, and Theorem 7.3 classifies the remaining squarefree semiprime. Content inflation preserves and reflects both closure criteria, by Section 4. Thus exactly the stated seed inflations remain. Their algebraicity and that of the other tuples follow from Sections 2–3. □

The proof uses the maximal-order and short-support method also present in [5], but does not assume its no-move theorem or claim that a profile-lowering move preserves Dec. In this subfamily we obtain the positive surface decomposition itself. Finite checks enumerate all mass-at-most-six split matrices of sizes 3×6 and 3×7, and all normalized potentials of norm at most six at six prime-power calibration levels, with direct age and surface-split checks. These verify the formulas rather than replace the proof for arbitrary exponents.

The inequalities used above do not cover $|K_5| \in \{4, 5\}$. We next classify these small shapes and their lower-order errors.

7.5. Arbitrary prime support at repeated three-primary depth

Lemma 7.9 (three slices). Let $C = \{1, 2, 3\} \times C'$, where C' is a product of sets of sizes at least four, with at most one size four and at most one size at most five. Suppose an integer split function $f \neq 0$ has mass at most six, and no three-point line carries three values of the same nonzero sign. Then, after permuting the three slices, one of the following holds:

1. One slice is a signed coordinate line of size four, five or six; the other two vanish.
2. For a coordinate line $L \subset C'$ of size $l \in \{4, 5\}$ and a nonempty proper subset $A \subset L$, the slices are $-\sigma\mathbf{1}_A, -\sigma\mathbf{1}_A, \sigma\mathbf{1}_{L\setminus A}$, where $\sigma = \pm 1$. Their mass is $l + |A|$, so $(l, |A|) = (4, 1), (4, 2), (5, 1)$.

Proof. Write f_i for the slices and $\mu_i = \|f_i\|_1$. Differences of slices are split on C'. If a slice vanishes, Lemma 7.5 makes each nonzero slice one signed line of mass at least four. Only one can occur, giving (1).

Otherwise $\mu_i \geq 1$, and each difference has mass at most five. By Lemma 7.5 a nonzero difference is one signed line of size four or five. There is at most one factor of size at most five, so all such lines have the same coordinate direction. Three pairwise different slices would give a sum of two signed parallel line indicators equal to a third one. Distinct parallel lines are disjoint, and equal lines give coefficients zero or two, so this is impossible. Two slices therefore agree, say with value h, and the third has value k, with $k - h = \sigma\mathbf{1}_L$. Off L, the no-same-sign condition forces $h = k = 0$. On L, it forces $(h, k) = (-\sigma, 0)$ or $(0, \sigma)$. This proves (2). If all slices agreed, their nonzero values would violate that condition. The empty product case likewise forces a same-sign three-line. □

Lemma 7.10 (surface depth). A pair-free balanced quadruple whose primitive conductor is divisible by nine is exactly a 3-standard quadruple $(y, y + m/3, y + 2m/3, -3y)$.

Proof. Divide out the content and put $3^a \parallel m$, $a \geq 2$. Choose a maximal order e containing 3^a, and use its odd multiplicity function on cosets of $H = \prod_{p|e} K_p$. Each coset has mass at most four, since $-1 \notin H$. Primitive Fourier nonvanishing gives the same split property as in Theorem 7.8. If no same-sign K_3-line exists, a zero slice would force one K_5-line of size four by Lemma 7.5. It consumes all four entries, whose sum modulo $e/5$ is $4w$, with w a unit; this is impossible since $9 \mid e/5$. If all three slices are nonzero, their differences have mass at most three and vanish by Lemma 7.5. Equal slices give a same-sign K_3-line after all. Thus the quadruple contains a complete 3-progression P_y, with $3y \neq 0$. Replacing it by $3y$ leaves a balanced pair, necessarily inverse. □

Theorem 7.11. At any odd primitive conductor divisible by nine, a residual sextuple exists only at conductor 45, and is the single degree-45 orbit in (3.1). There is no restriction on the number or exponents of the other prime divisors.

Proof. Let T be primitive, pair-free, outside $\mathrm{Dec} \cup \mathrm{Std}_5$, and put $3^a \parallel m$, $a \geq 2$. Choose a maximal order e containing 3^a. A same-sign K_3-line in its odd function would give a full positive 3-progression, hence a surface contraction as in Theorem 7.8. None exists. Lemma 7.9 applies on each nonzero coset, since $|K_3| = 3$ and the other factor sizes satisfy its hypotheses. A coset and its negative together account for as many entries of T as the mass on either coset.

In case (1), a size-six line must be a K_7-line. It consumes all six entries and their sum modulo $e/7$ is $6w$, impossible since $9 \mid e/7$. A size-five line is a full 5-progression, so the sixth entry makes T exactly 5-standard.

Consider case (2) in coordinates in $\mathbf{Z}/e$. Choose w on the distinguished row's K_5-line L, and parametrize the K_3-coordinate so that this third, distinguished slice has coordinate one. Thus the two other slices are the nonidentity K_3-translates of this row. All its points, including those of A, are w modulo $e/5$. The coordinate sum of a complete K_3-line through $u \in A$ is $3u$ (indeed $3uk \equiv 3u \pmod{e}$ for $k \in K_3$, so this holds for every point of the line). For $l = 4$, the sum on L is $5w - z$, where z is the unique missing element of the 5-progression and is divisible by 5; for $l = 5$ it is $5w$. Consequently the signed sum of the top entries is $\sigma(l - 3|A|)w$ modulo $e/5$. For $(l, |A|) = (4, 2), (5, 1)$, all six entries are tops and this sum is $\mp 2w$, impossible since $3 \mid e/5$. For $(4, 1)$, there are five tops. Their remaining entry lies in the subgroup of order e by zero sum; its residue modulo $e/5$ is $-\sigma w$, and its residue modulo 5 is $3\sigma u \neq 0$, where $A = \{u\}$. It too has order e. It cannot belong to the already specified coset pair; another pair would have mass one. A nonzero split function on $K_3 \times C'$ has mass at least three: either its three slices are nonzero, or its nonzero slices are split on C' and have mass at least four. This excludes the extra top.

Only case (1) with a size-four K_5-line remains. In the original group write

$$T = (P_x \setminus \{z\}) \sqcup \{b, c\}, \qquad P_x = \{x, x + m/5, \dots, x + 4m/5\}.$$

Here $5 \parallel e$, x has order e, and z has order $e/5$. The distribution identity makes $Q = (5x, b, c, -z)$ balanced. Its entries are nonzero. If Q has an inverse pair, it is a union of two inverse pairs. The pair $(5x, -z)$ would imply $20x = 0$, impossible at this depth; (b, c) is forbidden by pair-freeness. Any cross-pair gives $\{b, c\} = \{z, -5x\}$, making T standard.

Thus Q is pair-free. Its two entries $5x, -z$ have full 3-primary depth a. Lemma 7.10 makes Q 3-standard. Both entries must belong to its progression: its fourth entry $-3y$ has 3-primary depth at most $a - 1$, since y has depth at most a. Hence $3(5x + z) = 0$. Using $5z = 5x$ gives $90x = 0$, so $e \mid 90$. As $9 \mid e$, $5 \parallel e$ and e is odd, $e = 45$. The two progressions and their distinguished entries all lie in the subgroup of order 45, so T does too. Primitivity gives $m = 45$. It remains to identify T. Here x has order 45; after multiplication by a unit, $x = 1$ and $P_1 = (1, 10, 19, 28, 37)$. The entries 5 and $-z$ of the progression of Q are distinct, so $5 + z \in \{15, 30\}$; with $z \in P_1$ this gives $z = 10$. The progression of Q is therefore $(5, 20, 35)$, its fourth entry is $-15 = 30$, and $\{b, c\} = \{20, 30\}$. Thus $T = (1, 19, 20, 28, 30, 37)$, the degree-45 orbit in (3.1). It is not in Dec by Lemma 7.18, since $15x \neq 0$, and it is not 5-standard, because only four of its entries are congruent modulo 9. □

7.6. The remaining two-prime support

Theorem 7.12. For every $a, b \geq 1$, the residual orbits at degree $3^a 5^b$ are exactly the inflation of the degree-45 seed when $a \geq 2$, and the empty set when $a = 1$.

Proof. By content reduction, Theorem 7.11, Proposition 7.4 and Theorem 7.3 at degree 15, only primitive degree $m = 3Q$, $Q = 5^b$, $b \geq 2$, remains. Use $F(s, u)$ on $U_3 \times U_Q$, and $D(u) = F(1, u) - F(-1, u)$, as in Theorem 7.8. Primitive

Fourier vanishing makes D even and constant on the five-point fibres of $U_Q \to U_{Q/5}$. Its mass is at most twelve. A nonzero such function must therefore be

$$D = \sigma(\mathbf{1}_C + \mathbf{1}_{-C}), \qquad \sigma = \pm 1, \tag{7.11}$$

on one pair of distinct fibres, of total mass ten.

First suppose $D \neq 0$. Oddness of F gives $\|F\|_1 - \|D\|_1 \in 4\mathbf{Z}_{\geq 0}$ (with $\alpha = F(1,u)$, $\beta = F(-1,u)$, the contribution $|\alpha| + |\beta| - |\alpha - \beta|$ at u equals that at $-u$ and is even); since $\|F\|_1 \leq 12$, equality holds. The five unit entries all have the same 3-coordinate σ. Zero sum forces the sixth entry to have that same nonzero 3-coordinate, so there is no entry of order Q. Put $W(u) = F(1,u) + F(-1,u)$. It is odd, supported on $C \cup (-C)$, and equals ± 1 there. At a primitive odd character of Q, the balance equation gives the Fourier coefficient of W multiplied by $1 - \chi(3)$, by (7.10). This factor is nonzero: 3 generates U_{5^b}, since $3^4 = 1 + 16 \cdot 5$, and binomial induction gives order $4 \cdot 5^{b-1}$; in particular a power of 3 is -1, where an odd character has value -1. Thus W is fibre-constant as well. Equations (7.11) and the constant values of W force all five unit entries into one 5-progression. The sextuple is exactly 5-standard.

Now let $D = 0$, so $F(1,u) = F(-1,u) = v(u)$, with v integer and odd. Let $G(y) = n_{3y} - n_{-3y}$ on U_Q. Primitive Fourier vanishing makes

$$E = G + v - v(3\,\cdot) \tag{7.12}$$

odd and fibre-constant, with $\|E\|_1 \leq 2N_Q + 2N_U \leq 12$, where $N_U = \|v\|_1 = 2r$ and $N_Q = \|G\|_1/2$. Suppose $E \neq 0$. It equals $\sigma(\mathbf{1}_C - \mathbf{1}_{-C})$, of mass ten. On the lattice of integer odd functions, write $\epsilon(h) = \|h\|_1/2 \pmod 2$. This is additive, since it is the sum of the coefficients modulo two on any choice of one representative of each antipodal pair. Multiplication by 3 preserves ϵ. Hence $1 = \epsilon(E) = \epsilon(G) = N_Q \pmod 2$. Thus $N_U + N_Q$ is odd and at most six, while the norm bound forces it to be at least five. It equals five, and equality holds in the triangle inequality for (7.12). In particular both v and $v(3\,\cdot)$ are supported on $C \cup (-C)$. But this set is disjoint from its multiple by 3: such an intersection would require $3 \equiv \pm 1 \pmod{5^{b-1}}$, impossible. Therefore $v = 0$, $N_Q = 5$, and the sixth entry is also divisible by 3 by zero sum. This contradicts primitivity. We have proved $E = 0$.

Consequently the exact equations (7.8)–(7.9) hold with $Q = 5^b$. Their positive parts form a balanced block of length $2r + d \leq 6$, as in Theorem 7.8. Primitivity excludes $v = 0$. For $r = 1$ the block has length four and leaves a forbidden balanced pair. For $r = 3$, invariance would require $Q \mid 3^j - 1$ for $j \leq 3$, impossible at $Q \geq 25$. For $r = 2$, the same cancellation calculation forces $v = e_z + e_{3z}$, giving the explicit contraction of two 3-standard surface quadruples. This is Dec. Content reduction now proves the asserted equality of residual sets. □

It follows that every degree with at most two distinct odd prime divisors has the exact six-seed classification. At an arbitrary odd primitive conductor, the only remaining case is $3 \parallel m$ with at least three distinct prime divisors. Theorem 7.11 does not assert that residuals at a degree divisible by nine are absent: the degree-33, degree-39 or other seeds can still inflate there. Its conclusion concerns the primitive conductor.

Corollary 7.13 (computer verification through 2001). The six-seed equality holds for every odd $3 \leq m \leq 2001$. Theorem A contains this statement; the proof below is independent of Sections 7.5–7.8.

Proof. Section 4 handles $m \leq 599$. Above this bound, the proved infinite families leave exactly 103 odd degrees $601 \leq m \leq 2001$ divisible by 3 and supported on $\{3, 5\}$ (the four degrees 675, 1125, 1215, 1875) or on at least three primes (99 degrees), when the selection is made using the earlier Theorem 7.8 alone. The unchanged independent enumerator of the supplementary bundle enumerated all their nonpair Hodge orbits, including nonprimitive ones. Its output, checked by direct full-unit Python predicates, has precisely the predicted seed inflations at every degree. The other degrees are covered by Theorems 7.3, 7.6, 7.8 and Proposition 7.4. Theorem 7.12 is not needed to select or validate these 103 degrees. □

7.7. Arbitrary prime support without five

We use the short split-function distinction of [5, Step 5 of the odd-divisible-by-three subsection], now without assuming the absence of a lowering move. Its simplification when 5 is absent gives a positive classification on arbitrary prime support.

Lemma 7.14. Suppose $N > 1$ has all prime divisors at least seven. Let $h : U_N \to \mathbf{Z}$ have mass at most twelve and be split on cosets of $\prod_{p|N} K_p$, where $K_p = \ker(U_N \to U_{N/p})$. If h is even and nonzero, either N is a prime $q \in \{7, 11, 13\}$

and h is constant, or

$$h = \sigma(\mathbf{1}_L + \mathbf{1}_{-L}), \qquad L = wK_7, \quad 7 \parallel N, \quad N/7 > 1. \tag{7.13}$$

If h is odd and nonzero, its only possibility is the same line pair with a minus sign between the indicators.

Proof. For prime N, split means constant and the assertion follows from parity and $N - 1 \le 12$. If N is not squarefree, opposite cosets are distinct and each has mass at most six. Lemma 7.5 leaves only a size-six K_7-line, and its opposite already consumes all twelve units of mass.

If N is squarefree composite, one of its primes p is at least eleven. Slice along U_p, whose size is at least ten. Opposite slices have equal mass. If all slices are nonzero, each has mass at most two and one has mass one. Every difference from that slice has mass at most three, hence vanishes by Lemma 7.5. All slices would be the same signed atom. Its fixed coordinates in the other factors are distinct from their negatives, contradicting either parity. There is therefore a zero slice. Each other slice is split on the remaining factors and has mass at most six. Again Lemma 7.5 gives the stated pair of size-six lines. □

Theorem 7.15. If m is odd and $5 \nmid m$, its residual orbits are exactly the inflations of the degree-33 and two degree-39 seeds whose conductors divide m. There is no bound on prime support or exponents.

Proof. Reduce content, discard inverse pairs, and suppose that T is residual. Theorem 7.6 excludes $3 \nmid m$, and Theorem 7.11 excludes $9 \mid m$. Thus $3 \parallel m$.

Every maximal entry order is divisible by 3. Indeed, a maximal order e prime to three has all primes at least seven. Its odd function has no primitive Fourier modes, so Lemma 7.14 makes it a pair of K_7-lines. All six entries then have order e and lie on a positive line. Their sum modulo $e/7$ is $6w$, impossible because $e/7 > 1$ is coprime to six.

Fix a maximal order $e = 3N$. The case $N = 1$ is impossible: split on U_3 means constant, whereas its odd function is nonzero. If an entry has order f divisible by N, it divides a maximal order e' divisible by 3. Then $e \mid e'$, so maximality gives $e' = e$ and $f \in \{N, 3N\}$. Thus primitive character equations at N involve only these two strata. Use $F(s, u)$ and $D = F(1, \cdot) - F(-1, \cdot)$ as before; D is even and split.

Suppose $D \neq 0$ and N is composite. Lemma 7.14 gives (7.13), of mass twelve, so all six entries are tops at e, $m = e$, and their 3-coordinate is one common sign. Write $N = 7R$, where $R > 1$ has primes at least eleven. The other coordinates of the tops are $\varepsilon_k wk$, for $k \in U_7$, with $\varepsilon_k = \pm 1$. Zero sum modulo $R \ge 11$ forces three plus and three minus signs. Zero sum modulo 7 then makes the plus set $\{1, 2, 4\}$ or $\{3, 5, 6\}$, after factoring out w. Thus $\varepsilon_k = \tau\lambda_7(k)$, with λ_7 the odd quadratic character of U_7.

There is an even primitive character ψ of R with $\psi(3) \neq -1$. To see this, at each prime power factor choose the second power of a faithful character; replacing that choice at one factor by its fourth power preserves evenness and primitivity, and changes its value at 3, since $9 \neq 1$ there. For $\chi = \lambda_7\psi$, the primitive odd character equation at N has the nonzero factor $1 - \chi(3) = 1 + \psi(3)$. Its remaining sum is a nonzero scalar times $\sum_{k \in U_7} \varepsilon_k \lambda_7(k) = 6\tau$, a contradiction. This is the seven-line argument of [5, Step 7 of the odd-divisible-by-three subsection]; the equal sign counts here follow from zero sum and $R \ge 11$.

Next suppose $D = 0$, so $F(1, u) = F(-1, u) = v(u)$ is odd, and put $G(y) = n_{(m/N)y} - n_{-(m/N)y}$ on the order-N stratum. Formula (7.10), in the subgroup of order $3N$, makes $E = G + v - v(3\,\cdot)$ odd with zero primitive Fourier modes. Its mass is at most $2(N_{3N} + N_N) \le 12$. If $E \neq 0$, Lemma 7.14 makes it a seven-line pair of mass twelve. Equality then holds in the triangle inequality, so v and $v(3\,\cdot)$ are both supported on $L \cup (-L)$. This set and its multiple by 3 are disjoint, because $N/7 \ge 11$ cannot divide $3 - 1$ or $3 + 1$. Hence $v = 0$, contradicting the presence of tops at $3N$. Therefore $E = 0$.

The positive balanced block now has length $2r + d \le 6$, with $r = \|v\|_1/2 \ge 1$. The argument of Theorem 7.8 excludes $r = 1$ and makes $r = 2$ a surface contraction; its invariance step uses only that a positive support of two points stable under multiplication by 3 would force $N \mid 3^j - 1$ with $j \le 2$, impossible for $N \ge 7$. For $r = 3$ it requires $N \mid 3^j - 1$ for some $j \le 3$. As the primes of N are at least seven, its only possibility is $N = 13$. The positive support of v is then one multiplication-by-3 orbit of length three. Up to units the sextuple is $(1, 14, 16, 22, 29, 35)$ at 39. It exhausts T, so its primitive conductor is 39.

It remains that every maximal order has nonzero D and $N = q \in \{7, 11, 13\}$. Their numbers of tops are at least $(q-1)/2$, respectively three, five and six. Two distinct such maximal orders would require at least eight entries. There is only

one, and primitivity gives $m = 3q$. The complete semiprime classification in Theorem 7.3 finishes the proof. Inflation both preserves and reflects the positive predicates. □

Theorems 7.11, 7.12 and 7.15 leave the primitive conductors satisfying $3 \parallel m$, $5 \mid m$, with at least three distinct prime divisors. The next subsection completes the positive classification on this domain as well.

7.8. Positive classification at arbitrary odd conductor

The proof treats the remaining primitive conductors, those with $3 \parallel m$, $5 \mid m$ and at least three distinct prime divisors. Its inputs are Theorem 7.6, the repeated-three reduction in Theorem 7.11 and the two-prime results above; no finite census is used. We first record the small-kernel classification needed for the remaining case. As in [5, Proposition 3.18 and Lemma 3.31], these are statements about split functions before a no-move assumption is imposed. The dependence is sequential: Lemma 7.16 classifies the Fourier kernels; Lemma 7.17 uses it to classify surfaces; Lemma 7.18 applies that surface classification to a partial five-progression; and Lemma 7.19 proves equality of the entire lower-stratum functions. Theorem 7.20 then treats the nonzero row difference. None of these lemmas assumes Theorem A or any finite census.

Lemma 7.16 (short kernels prime to six). Let $N > 1$ be prime to six, and let $h : U_N \to \mathbf{Z}$ be nonzero, split on the cosets of $\prod_{p|N} K_p$, of mass at most twelve. In the line formulas below $\sigma \in \{1, -1\}$.

If h is odd, either

$$h = \sigma(\mathbf{1}_L - \mathbf{1}_{-L}), \quad L = wK_p, \quad |K_p| \in \{4, 5, 6\}, \quad L \cap (-L) = \varnothing, \tag{7.14}$$

or $N = 35$ and $\|h\|_1 = 12$. If h is even, either it has the analogous form with a plus sign, or $N = q \in \{5, 7, 11, 13\}$ and h is constant, or $N = 35$. In the last even case, unless it is already a line pair, its mass is twelve and it has eight values of one sign and four of the other, all of absolute value one.

Proof. At prime N, split means constant. Oddness excludes this case for an odd function; for an even function its mass is $(N-1)|h(1)| \le 12$, giving exactly the stated prime possibilities.

If N is not squarefree, choose $p^a \parallel N$, $a \ge 2$. Every element of $H = \prod K_p$ is one modulo p^{a-1}, whereas -1 is not. Thus a coset and its negative are distinct. Parity pairs their masses, so each has mass at most six. The kernel factor sizes are p for repeated primes and $p - 1$ otherwise; all are at least four and only K_5 can have size four. Lemma 7.5 gives precisely one signed line in each nonzero coset. That line has mass at least four. Its opposite consumes at least four more, so no second pair of nonzero cosets fits in mass twelve. This proves (7.14) and its even counterpart for every exponent pattern.

If N is squarefree composite with a prime $p \ge 11$, slice along U_p, of size at least ten. Opposite slices have equal mass. If all are nonzero, they have mass at most two, with at least one of mass one. All their differences from that slice have mass at most three and vanish by Lemma 7.5. All slices would equal one signed atom at the same point in the other coordinates. That point differs from its negative because at least one other odd prime factor is present. This contradicts either parity. A zero slice therefore exists. Every nonzero slice is split in the other factors, of mass at most six, hence one signed line. Its opposite already consumes at least eight units; no other pair of slices can occur.

The only remaining squarefree composite is 35: any squarefree integer prime to six with at least three prime factors has a factor at least eleven. For odd h, its matrix on $U_5 \times U_7$ is $u(a) + v(b)$, with odd potentials simultaneously integral or half-integral. Formula (7.6) shows that two nonzero integral potentials cost at least $2(2 + 3 - 1) = 8$ positive entries, exceeding six: one nonzero antipodal value in each potential already occupies one row and one column of the 2×3 array of maxima. One integral potential gives a single line pair. Half-integral potentials cost at least six positive entries, so have exactly mass twelve.

For even h, both potentials may be chosen even: their differences under negation are equal constants, which change sign under negation and hence vanish. Its values form an additive 2×3 integer matrix M, each value repeated four times, with $\sum |M_{ij}| \le 3$. Its row difference is an integer δ. If $\delta = 0$, there is one signed nonzero column, a K_5-line pair. If $|\delta| \ge 2$, the three columns already cost at least six, impossible. For $|\delta| = 1$, each column is $(\delta, 0)$ or $(0, -\delta)$. Equal columns give a K_7-line pair; otherwise the nonzero signs occur on eight and four points. □

We fix the row and stratum conventions for the next three arguments. Suppose $3 \parallel m$, all maximal orders are divisible by three, and $e = 3N$ is maximal. The order $e = 3$ cannot occur: its only odd character is primitive, and (7.7a) would

force its nonzero odd multiplicity function to vanish. Thus $N > 1$ and $\gcd(N, 6) = 1$. If an entry order f is divisible by N, choose a maximal order e' divisible by f. Since $3 \mid e'$, we have $3N \mid e'$, whence $e' = e$. It follows that $f \in \{N, 3N\}$. This proves the two-stratum isolation used below.

On the order-$3N$ stratum identify an entry with $(\sigma, u) \in U_3 \times U_N$ through the subgroup embedding $z \mapsto (m/(3N))z$, and write its odd multiplicity as F. Put $D(u) = F(1, u) - F(-1, u)$ and $W(u) = F(1, u) + F(-1, u)$. Oddness of F makes D even and W odd. The primitive odd characters at $3N$ are the product of the nontrivial character at 3 with an even primitive character at N. Equation (7.7a) therefore kills every primitive Fourier coefficient of D, so D is split on the kernel-product cosets. If N_e denotes the number of entries of order e, pair-freeness gives $\|F\|_1 = 2N_e$. Directly on each pair $\{u, -u\}$,

$$\|F\|_1 - \|D\|_1 \in 4\mathbf{Z}_{\geq 0}, \qquad \sum_{\mathrm{ord}(a)=3N} \sigma_a = \frac{1}{2} \sum_{u \in U_N} D(u). \tag{7.14a}$$

Here $\sigma_a \in U_3$ is the 3-coordinate of the entry a. The first assertion follows from $|\alpha| + |\beta| - |\alpha - \beta| \in 2\mathbf{Z}_{\geq 0}$ for $\alpha = F(1, u)$, $\beta = F(-1, u)$, with an additional factor two from $u, -u$; the second is the count of the order-$3N$ entries by their 3-coordinates.

If $D = 0$, write the two equal rows as the integer odd function υ, and let $G(y) = n_{(m/N)y} - n_{-(m/N)y}$ be the order-N odd function. Equation (7.10) and (7.7a), with conductor N, give

$$E = G + \upsilon - \upsilon(3\,\cdot), \qquad \widehat{E}(\chi) = 0 \quad (\chi \text{ primitive modulo } N), \tag{7.14b}$$

where $\widehat{E}(\chi) = \sum_{u \in U_N} E(u)\overline{\chi(u)}$. Oddness supplies the even-character vanishings, so Lemma 7.16 applies to E. Put $B_0 = N_{3N} + N_N$. Then

$$N_{3N} = \|\upsilon\|_1 \in 2\mathbf{Z}, \qquad \|E\|_1 \leq \|G\|_1 + 2\|\upsilon\|_1 = 2B_0. \tag{7.14c}$$

There are two distinct uses of this formula. If the entire tuple lies in these strata, (7.10) gives, for every $t \in U_N$,

$$0 = \frac{1}{2} \sum_{u \in U_N} E(u) b_N(tu). \tag{7.14d}$$

Thus the positive part of E is itself balanced, since $b_N(-x) = -b_N(x)$. If instead $E = 0$ is proved as an equality of functions, the positive parts of F, G form a balanced submultiset of the original tuple. We do not infer (7.14d) merely from vanishing of primitive Fourier coefficients.

Lemma 7.17 (odd surface classification). A pair-free balanced quadruple at odd degree is either 3-standard or has primitive conductor 15 or 21.

Proof. Reduce content. If $3 \nmid m$, [1, Theorem A, $n = 2$] makes the quadruple a union of inverse pairs. If $9 \mid m$, use Lemma 7.10. Assume $3 \parallel m$. A maximal order e prime to three has an odd split function of mass at most eight. Lemma 7.16 makes it a K_5-line pair of mass eight. Its four positive entries have sum $4w$ modulo $e/5 > 1$, impossible as $e/5$ is prime to six. All maximal orders are therefore divisible by three.

Fix one such order $3N$. As in Theorem 7.15, only the order strata $3N, N$ can contribute at primitive characters of N. If its even row difference D vanishes, put $F(1, u) = F(-1, u) = \upsilon(u)$ and $E = G + \upsilon - \upsilon(3\,\cdot)$. The odd error has mass at most eight. If nonzero, Lemma 7.16 makes it a four-point K_5-line pair and forces equality in the norm bound. Both υ and $\upsilon(3\,\cdot)$ are then supported on $L \cup (-L)$, disjoint from its multiple by 3 because $N/5 > 1$ is prime to six. This contradicts $\upsilon \neq 0$. Thus $E = 0$. The positive balanced block has length $2r + d \leq 4$. The case $r = 1$ is a 3-standard quadruple exhausting the original one; $r = 2, d = 0$ would require $N \mid 3^j - 1$ with $j \leq 2$, impossible at odd $N \geq 5$.

If $D \neq 0$, its mass is at most eight. Lemma 7.16 leaves either a four-point K_5-line pair, or a constant at $N = 5, 7$. The line-pair case uses all four tops, with the same 3-coordinate, contradicting zero sum modulo 3. Thus each maximal order is 15 or 21, requiring respectively at least two or three tops. They cannot coexist in a quadruple. Primitivity makes the conductor 15 or 21. □

Lemma 7.18 (four points of a five-progression). If a primitive pair-free residual sextuple contains four distinct points of $P_x = \{x, x + m/5, \ldots, x + 4m/5\}$, with $5x \neq 0$, then m divides 45 or 105; if $m = 105$, it is one of the two degree-105 orbits in (3.1). Conversely, a pair-free sextuple that contains four distinct points of P_x, with $15x \neq 0$, is not in Dec.

Proof. Five points would make the tuple 5-standard. Otherwise write $T = (P_x \setminus \{z\}) \sqcup \{b, c\}$. The quadruple $Q = (5x, b, c, -z)$ is balanced, because $\sum_{y \in P_x} b_m(ty) = b_m(5tx)$ (distribution identity over the five lifts of x mod $m/5$) and

$b_m(-tz) = -b_m(tz)$. Its entries are nonzero: $5x \neq 0$, and $z = 0$ would give $5x = 5z = 0$. It is pair-free. The pair $\{b, c\}$ is excluded because T is pair-free. The pair $\{5x, -z\}$ would give $z = 5x$, hence $5x = 5z = 25x$, $20x = 0$ and $5x = 0$ at odd degree. A pair $\{b, -z\}$ or $\{c, -z\}$ would put z back in T, giving five points of P_x. A pair $\{5x, b\}$ leaves the balanced pair $(c, -z)$, so $c = z$, again five points; $\{5x, c\}$ is symmetric.

If Q has primitive conductor $f_Q = 15$ or 21, all its entries lie in the subgroup of order f_Q. Since $x = z - jm/5$ for some j, every entry of T lies in the subgroup of order $\operatorname{lcm}(f_Q, 5)$. Primitivity gives $m \mid 15$ or $m \mid 105$.

Otherwise Lemma 7.17 makes Q 3-standard. If $5x, -z$ both lie in its progression, then $3(5x + z) = 0$, and $5z = 5x$ gives $90x = 0$. If $5x$ is its distinguished fourth entry, the same identities give $3z = 5x$, hence $10x = 0$, impossible since $5x \neq 0$. If $-z$ is the distinguished entry, they give $z = 15x$, hence $70x = 0$. In the possible cases all entries are generated by $x, m/3, m/5$. Their group has order dividing 45 or 105, respectively. Primitivity proves the first assertion.

Now let $m = 105$. The first 3-standard case is excluded by primitivity, since its group has order dividing 15, and the second is impossible. If Q is not 3-standard, Lemma 7.17 gives $f_Q = 15$ or 21, and primitivity excludes 15. Then $Q = 5Q_0$ for a pair-free balanced quadruple Q_0 at degree 21 which is not 3-standard. Apply the analysis of Theorem 7.3 at $(3, 7)$ to Q_0, with total length four. If $\lambda = 0$, then $N_U \geq 6|\mu|$ forces $\mu = 0$ and $r = 1$, which makes Q_0 3-standard. Hence $\lambda = 1$, and $N_U \geq 3$, $d(\upsilon) \geq 1$ force $N_U = 3$, $d(\upsilon) = 1$: the values of υ along the cycle of 3 are $\pm(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}, -\frac{1}{2})$ up to cyclic shift, and Q_0 lies in the unit orbit of $(1, 10, 15, 16)$. Write $5x = 5\alpha$ and $-z = 5\zeta$ with α, ζ distinct entries of Q_0. As $z \in P_x$ means $z \equiv x \pmod{21}$, we get $\alpha + 5\zeta \equiv 0 \pmod{21}$. After a unit, $Q_0 = (1, 10, 15, 16)$, and the only solution is $(\alpha, \zeta) = (16, 1)$. Thus P_x is the set of lifts of 16 modulo 21, $z = 100$, $\{b, c\} = \{50, 75\}$, and $T = (16, 37, 50, 58, 75, 79) = 16 \cdot (1, 22, 43, 64, 90, 95)$. In the 3-standard case $z = 15x$, and $z \equiv x \pmod{21}$ gives $3 \mid x$; $5x \neq 0$ gives $7 \nmid x$. The data P_x, z and Q depend only on x mod 21, so after a unit $x = 3$, $z = 45$, $Q = (15, 50, 85, 60)$ and $T = (3, 24, 50, 66, 85, 87)$.

For the converse, suppose $T = B \sqcup C$ is a surface contraction. Since T is pair-free, both quadruples $Q_1 = (B, -k)$ and $Q_2 = (C, k)$, with k the distinguished sum, are pair-free, because a balanced quadruple containing an inverse pair is a union of two inverse pairs. One of B, C, say B, contains two of the four points of P_x, whose difference has order five. Every point x' of P_x satisfies $15x' = 15x \neq 0$. By Lemma 7.17, Q_1 is 3-standard or has primitive conductor 15 or 21. The subgroup of order 21 contains no element of order five, and the subgroup of order 15 is killed by 15, contrary to $15x' \neq 0$. If $Q_1 = \sigma_{3,y}$, two progression entries differ by a multiple of $m/3$, so the two points are $-3y$ and $y + jm/3$. Their difference $4y + jm/3$ has order five, whence $20y \equiv 0 \pmod{m/3}$, $5y \equiv 0 \pmod{m/3}$ as m is odd, and $15y = 0$. Then 15 kills every entry of Q_1, again contradicting $15x' \neq 0$. □

Lemma 7.19 (zero row difference). Suppose T is a primitive pair-free balanced sextuple at $3 \parallel m$, and every maximal entry order is divisible by three. If its row difference vanishes at one maximal order $3N$, then $T \in \mathrm{Dec}$ or its primitive conductor is 39 and it is the second degree-39 seed.

Proof. Only orders $3N, N$ contribute at conductor N, by the isolation argument above. Use υ, G, E in (7.14b), with $\upsilon \neq 0$ since this maximal stratum is present, and put $B_0 = N_{3N} + N_N \leq 6$. For an integer odd function define $\epsilon(h) = \|h\|_1/2 \pmod 2$. Choosing one representative of each antipodal pair expresses it as the sum of the coefficients modulo two. It is therefore additive and is preserved by multiplication by 3. Consequently

$$\epsilon(E) = \epsilon(G) = N_N \pmod 2.$$

Since $N_{3N} = \|\upsilon\|_1$ is even, this is exactly $B_0 \equiv \|E\|_1/2 \pmod 2$. Lemma 7.16 and (7.14c) leave masses eight, ten or twelve for a nonzero E.

At mass eight, either $B_0 = 4$, giving equality in the norm bound, or $B_0 = 6$. In the equality case the support of each summand of $E = G + \upsilon - \upsilon(3\,\cdot)$ is contained in $S = L \cup (-L)$, where L is a four-point K_5-line. Thus $\operatorname{supp}(\upsilon) \subset S \cap 3S$. But points of S are congruent to $\pm w \pmod{N/5}$, whereas points of $3S$ are congruent to $\pm 3w$. An intersection would force $N/5 \mid 2$ or $N/5 \mid 4$, impossible for $N/5 > 1$ prime to six. This contradicts $\upsilon \neq 0$. If $B_0 = 6$, the entire sextuple lies in the two strata, so (7.14d) makes the positive part of E a balanced quadruple entirely in U_N, pair-free because E is odd. This contradicts [1, Theorem A, $n = 2$]. At mass ten, parity forces $B_0 = 5$, again giving equality and the same support contradiction; here $5^2 \mid N$, so $N/5 \geq 5$. This argument does not require all entries to lie in the two strata. At mass twelve, $B_0 = 6$, and (7.14d) makes the positive part of E a balanced all-unit sextuple at N, pair-free because E is odd. Theorem 7.6 would make it 5-standard, whose entry $-5x$ is a nonunit; this too is impossible. Therefore $E = 0$.

Since $E = 0$, the positive parts form a balanced block $B \subseteq T$. Set $r = \|v\|_1/2 \geq 1$ and $d = \|v(3\cdot) - v\|_1/2$. Then $|B| = 2r + d \leq 6$. Write $e_z = \mathbf{1}_z - \mathbf{1}_{-z}$.

If $r = 1$, then $v = e_z$. As $N \geq 5$ and $3 \not\equiv \pm 1 \pmod N$, we have $d = 2$ and $|B| = 4$. Its complement is a balanced pair and hence an inverse pair, contradicting pair-freeness of T.

If $r = 2$, a repeated atom $2e_z$ gives $d = 4$, too large. Otherwise $v = e_z + e_w$, with $z \neq \pm w$. Invariance under multiplication by 3 would require a positive orbit of length at most two, forcing $N \mid 3^j - 1$ for $j \leq 2$, impossible for $N > 1$ prime to six. Thus $0 < d \leq 2$. In

$$v(3\cdot) - v = e_{3^{-1}z} + e_{3^{-1}w} - e_z - e_w$$

the norm can drop below eight only by cancellation of two oppositely signed atoms. Apart from the excluded self coincidences, this requires $w = 3z$ or $z = 3w$; negative coincidences reinforce rather than cancel. Relabel to obtain $v = e_z + e_{3z}$. Now $G = e_{3^{-1}z} - e_{3z}$ has half-norm two: the two residues are distinct, since equality would imply $N \mid 8$. If they are negatives of each other, then $N \mid 10$, so $N = 5$ and $G = 2e_{3^{-1}z}$ still has half-norm two. The two order-N entries then coincide, which is allowed since entries may repeat. Hence $d = 2$. Take 3-standard surface quadruples with parameters congruent to $z, 3z \pmod N$, in the subgroup of order $3N$. The distinguished entry of the first and the progression entry divisible by 3 of the second are inverse. Their contraction has the prescribed top rows and lower function $G = v(3\cdot) - v$. More explicitly, choose $x \equiv z \pmod N$ in $\mathbf{Z}/(3N)$ and put $\kappa = m/(3N)$. The two surfaces are $\sigma_{3,x}$ and $\sigma_{3,3x}$ at degree $3N$. Remove their entries $-3x, 3x$. The remaining multiset is

$$T = \kappa\{x, x+N, x+2N, 3x+N, 3x+2N, -9x\}. \tag{7.14e}$$

Each parameter is a unit modulo N, so both standard tuples are admissible and the removed entry is nonzero. Their remaining triples have respective sums $3x, -3x$, precisely the nonzero distinguished sums in the definition of Dec. The top rows are $e_z + e_{3z}$, and the lower function is $e_{3^{-1}z} - e_{3z}$. These functions already have positive mass six, so no further inverse cancellation occurs. The multiset is therefore T, proving $T \in \mathrm{Dec}$.

If $r = 3$, then $d = 0$. The positive support of v is a union of multiplication-by-3 orbits of total multiplicity three. Every point is a unit, so an orbit of size $j \leq 3$ implies $N \mid 3^j - 1$. The only divisor greater than one prime to six among these three integers is 13. Hence $N = 13$, and the positive support is one orbit of length three. Up to units, take $v = e_1 + e_3 + e_9$; its two equal positive rows give $(1, 14, 16, 22, 29, 35)$ at 39. This block exhausts T, so primitivity makes its conductor 39. □

Theorem 7.20 (unrestricted positive classification). For every odd degree, the residual orbits outside $\mathrm{Dec} \cup \mathrm{Std}_5$ are exactly the inflations of the six seeds in (3.1).

We shall use one elementary local character fact in its proof. At an odd prime power p^s, take a faithful character θ of the cyclic unit group. For $p \geq 7$, the powers θ, θ^3 are two primitive odd characters, and θ^2, θ^4 are two primitive even characters. The same is true at 5^s for $s \geq 2$. Indeed, at prime level these powers are nonprincipal; at higher level their exponents are prime to p, so restriction to the last p-kernel is nontrivial. The two values at 3 in either pair differ: equality would give $\theta(9) = 1$, whence $9 = 1 \pmod{p^s}$, which is impossible. At 5^1 there is just one primitive even character, the quadratic character, whose value at 3 is -1. Products of the local primitive characters are primitive at the product modulus. In particular, at a modulus greater than one whose primes are at least seven, either prescribed parity has at least two possible values at 3.

Proof. Content reduction reflects both predicates, so it suffices to determine the primitive residual orbits. Theorems 7.3, 7.6, 7.8, 7.11, 7.12 and 7.15 and Proposition 7.4 settle every primitive conductor with at most two distinct prime divisors, every conductor prime to five and every conductor divisible by nine. Let T be a primitive residual at any remaining degree m. Then $3 \parallel m$, $5 \mid m$, and m has at least three distinct prime divisors; in particular $m \geq 105$ and $m \nmid 45$. By Lemma 7.18, if T contains four points of a 5-progression with nonzero distinguished entry, then $m = 105$ and T is one of the two degree-105 orbits in (3.1). We show that every other configuration is impossible.

Every maximal order is divisible by three. At a maximal order e prime to three, Lemma 7.16 gives either a positive K_5-line of length four or five, or mass twelve (a K_7-line pair or the exceptional case $e = 35$). Four points are the case just settled, and five points make T exactly 5-standard by zero sum. The full-mass alternative consumes the entire sextuple in the subgroup of order e, so primitivity would give $m = e$, contradicting $3 \mid m$.

For a maximal order $e = 3N$, Lemma 7.19 excludes $D = 0$. Apply the even part of Lemma 7.16 to its nonzero D. Equations (7.14a) give the possible top counts and their sum modulo 3.

A K_7-line pair has mass twelve, so all six entries are tops, and $m = 3N$. Write $N = 7R$, $R > 1$ prime to 42. Their other coordinates are $\varepsilon_k wk$, $k \in U_7$. Zero sum modulo R, which is prime to six, forces three plus and three minus signs: the possible differences are $-6, -4, -2, 0, 2, 4, 6$. Zero sum modulo 7 then gives $\varepsilon_k = \tau\lambda_7(k)$. If $R = 5$, so that $m = 105$, apply (7.7a) at the conductor 7 with $\chi = \lambda_7$. All six entries have order 105, and the Euler factor is $(1-\lambda_7(3))(1-\lambda_7(5)) = 4$, since 3 and 5 are not squares modulo 7. As λ_7 is odd, $\lambda_7(\varepsilon_k wk) = \tau\lambda_7(w)$ for every k, and the relation reads $24\tau\lambda_7(w)/\varphi(105) = 0$, a contradiction. If $R \neq 5$, there is an even primitive ψ on R with $\psi(3) \neq -1$. Indeed choose even primitive characters at its prime powers; at any factor p^s other than 5^1, that is, with $p \neq 5$ or $s \geq 2$, the second and fourth powers of a faithful character remain primitive and give different values at 3. The primitive character $\lambda_7\psi$ gives the same nonzero relation as in Theorem 7.15, a contradiction. The exceptional even matrix at $N = 35$ also has mass twelve; all six entries are then tops and $m = 105$. Each entry is congruent modulo 3 to its 3-coordinate σ_a, and by (7.14a) $\sum_a \sigma_a = \frac{1}{2}\sum_u D(u) = \frac{1}{2}(8-4)(\pm 1) = \pm 2$. This contradicts the zero sum of T modulo 3.

Consider a K_5-line pair of mass ten, where $5^2 \mid N$. There are exactly five tops, all with the same 3-coordinate σ. Zero sum puts the sixth entry in the subgroup of order $3N$, so primitivity gives $m = 3N$. The remaining entry also has 3-coordinate σ, so no entry has order N, and the only entries whose orders are divisible by N are the five tops. Write their other coordinates as $\varepsilon_j wk_j$, $j = 0, \ldots, 4$, on the five-point line. A primitive character at N restricts nontrivially to K_5. Unless all ε_j agree, $\sum_j \varepsilon_j \zeta_5^{aj} \neq 0$, with $\zeta_5 = e^{2\pi i/5}$, for every $a \not\equiv 0 \pmod 5$, by the minimal polynomial $1 + X + \cdots + X^4$. Choose a primitive odd character χ of N with $\chi(3) \neq 1$. Such a choice exists by varying the character at the repeated 5-factor between powers 1 and 3, or 2 and 4, of a faithful character, keeping the required parity; the values differ since $9 \neq 1$ modulo that factor. Its conductor-N balance equation is then nonzero, impossible. All signs therefore agree, giving a full positive 5-progression and hence a 5-standard sextuple.

The remaining composite possibility is a K_5-line pair of mass eight, with $N = 5R$, $R > 1$ prime to 30. It has four or six tops. Six would give their sum $4\sigma \neq 0 \pmod 3$, so there are four. No distinct maximal order can coexist: another composite needs at least four tops; a prime case $3q$, $q \neq 5$, needs at least three; and 15 already divides this order. Thus $m = 3N$, and there are two other entries. At most one has order N: the four tops contribute 4σ to the 3-coordinate sum, so the other two 3-coordinates sum to $-\sigma$ modulo 3, and two zero coordinates are impossible.

Write the four tops as $(\sigma, \varepsilon_k wk)$, $k \in U_5$, and put $A(\alpha) = \sum_{k \in U_5} \varepsilon_k \overline{\alpha(k)}$. For a primitive odd $\chi = \alpha\psi$ at N, with $\alpha \neq 1$ modulo 5 and ψ primitive modulo R, its equation is

$$(1 - \chi(3))\overline{\chi(w)}A(\alpha) + 2 \sum_{\mathrm{ord}(b)=N} \overline{\chi(z_b)} = 0. \tag{7.15}$$

Indeed (7.7a) involves only orders $3N, N$. Their denominators are $2\varphi(N), \varphi(N)$; the former also has the Euler factor $1 - \chi(3)$. Multiply by $2\varphi(N)$, and use $\overline{\chi(\varepsilon_k wk)} = \varepsilon_k \overline{\chi(w)\alpha(k)}$. This explains both the factor two and the absence of a 3-coordinate sign in (7.15). If there is no order-N entry and the signs are not constant, choose α with $A(\alpha) \neq 0$. Primitive characters of R of either prescribed parity take at least two different values at 3: at a prime power factor, the first and third faithful powers, or the second and fourth, are primitive and have different values. Choose ψ with $\psi(3) \neq \alpha(3)^{-1}$. Equation (7.15) fails.

If there is one order-N entry, a two-plus/two-minus sign pattern has $A(\alpha) = 0$ at some nontrivial α, again contradicting (7.15). Indeed, write the two positive positions as g^i, g^j for a generator of U_5. If $i - j$ is odd, the quadratic character has opposite values there; if $i - j = 2$, either quartic character does. The full nontrivial character sum is zero, so in both cases A is twice this zero two-term sum. For a three-against-one pattern with exceptional position k_0, $A(\alpha) = \pm 2\overline{\alpha(k_0)}$, and hence $|A(\alpha)| = 2$ for every nontrivial α. Fix an even primitive ψ and use the two quartic characters modulo 5, whose values at 3 are $i, -i$. Equation (7.15) would require both $|1 - i\psi(3)| = |1 + i\psi(3)| = 1$, whereas the sum of their squares is four. Thus the signs are constant in every possible case. The four tops are four points of one 5-progression, the case settled by Lemma 7.18.

Only prime cases $N = q \in \{5, 7, 11, 13\}$ remain. Their minimal top counts are respectively two, three, five and six. A single maximal order would give $m = 3q$, which has two prime divisors. Every pair of distinct maximal orders other than 15 and 21 requires at least seven entries. Suppose, finally, that 15 and 21 are both maximal, so that $m = 105$ and every entry order divides 15 or 21. Let $q \in \{5, 7\}$. The row difference D at $3q$ is a nonzero constant c_q, and

$N_{3q} \geq \frac{1}{2}(q-1)|c_q|$. If $|c_q| \geq 2$, then $N_{15} + N_{21} \geq 7$. Otherwise $D = \pm 1$ everywhere, and $W = F(1,\cdot) + F(-1,\cdot) \equiv D \pmod 2$ is an odd function on U_q without zeros. With $\widehat{W}(\chi) = \sum_u W(u)\overline{\chi(u)}$, some odd character χ modulo q has $\widehat{W}(\chi) \neq 0$, and $\chi(3) \neq 1$, because 3 generates U_q and $\chi(-1) = -1$. The entries whose orders are divisible by q have orders $3q$ and q; the tops contribute $\frac{1}{2}\widehat{W}(\chi)$ to the character sum, as χ is odd. Multiplying (7.7a) at the conductor q by $\varphi(3q) = 2\varphi(q)$ gives

$$\tfrac{1}{2}(1 - \chi(3))\widehat{W}(\chi) + 2 \sum_{\operatorname{ord}(b)=q} \overline{\chi(z_b)} = 0.$$

Hence $N_q \geq 1$. The strata $\{15, 5\}$ and $\{21, 7\}$ are disjoint, so T would have at least $(2 + 1) + (3 + 1) = 7$ entries, a contradiction.

Thus the primitive residuals are among the six seeds. Conversely, the six seeds are pair-free by inspection and balanced. At 33 and 39 their odd functions are given by (7.2)–(7.3) in Theorem 7.3; substituting (7.2)–(7.3) into the balance equations and applying (7.4), as in the proof of Lemma 7.1, makes every correlation vanish. At 45 and 105, (3.4)–(3.5) express each seed, after adding two inverse pairs, as the union of a 5-standard tuple and a surface Hodge tuple. The seeds lie outside $\mathrm{Dec} \cup \mathrm{Std}_5$: at 33 and 39 by Theorem 7.3, at 45 by Theorem 7.11, and at 105 by Lemma 7.18, since $15x \neq 0$ for both progressions and only four entries of either tuple are congruent modulo 21. The two degree-105 orbits are distinct, since the first has four unit entries and the second none. Reflection of both positive predicates shows that every listed seed inflation remains residual; surjectivity on units preserves the distinction between the two seeds at each common conductor. □

The proof uses neither a bounded conductor hypothesis nor a premise that a profile-lowering move preserves positive decomposition. The four-against-one and three-against-one distinctions are resolved in the positive tuple itself. The finite enumerations of Section 4 and Corollary 7.13 are independent checks of the resulting classification.

Proposition 7.21 (independence from the finite census). The proof of Theorem A in Sections 2, 3 and 7 uses no computer enumeration; in particular it does not use Proposition 4.1.

Proof. By content reduction and Lemma 2.1 it suffices to determine the primitive residual orbits at each odd conductor m. We list where each case is settled.

(a) $\gcd(m, 6) = 1$: Theorem 7.6. Its proof uses Theorem 7.3 only at $(p, q) = (5, 7)$, whose case $p = 5$ is settled by Lemma 7.2 and the argument following it.

(b) m a prime power: Proposition 7.4, whose prime case is [1, Lemma 2.3].

(c) $9 \mid m$: Theorem 7.11, which identifies the degree-45 seed and uses Lemma 7.18 for its non-decomposability.

(d) $m = 3^a 5^b$ with $a = 1$: Theorem 7.12, through Theorem 7.3 at 15.

(e) $m = 3^a q^b$ with $q \geq 7$: Theorem 1.3, through Proposition 7.4, Theorem 7.8 and Theorem 7.3, which treats the degrees 15, 21, 33, 39 by the potential equations (7.6a).

(f) $5 \nmid m$, $3 \parallel m$, at least three prime divisors: Theorem 7.15. Its exits through Theorem 7.3 at $m = 3q$, $q \in \{7, 11, 13\}$, and through the second degree-39 seed have two prime divisors, so they do not occur under these hypotheses.

(g) $3 \parallel m$, $5 \mid m$, at least three prime divisors: Theorem 7.20. Its exits are a 5-standard tuple; Lemma 7.18, giving $m \mid 45$ or $m \mid 105$; Lemma 7.19, giving Dec or conductor 39; and the prime cases, which give $m = 3q \leq 39$, at least seven entries, or, when 15 and 21 are both maximal, a contradiction by (7.7a). Under these hypotheses the only conductor reached is $m = 105$, where Lemma 7.18 identifies the two seeds.

The cases (a)–(g) exhaust the odd conductors. From Section 4 the cited results use only content reduction and the reflection of both predicates under inflation, which involve no computation; none uses Proposition 4.1 or another census. □

The proof dependencies in Section 7 are summarized below, where each arrow points from a proved input to the result whose proof invokes it and (7.7a) denotes the conductor Fourier relation in Section 7.4:

- Lemmas 7.1 and 7.2 → Theorem 1.2 and Theorem 7.3 ($m = pq$).
- Lemma 7.5 and Theorem 7.3 (at $m = 35$) → Theorem 7.6 ($\gcd(m, 6) = 1$).
- Lemma 7.7 and (7.7a) → Theorem 7.8 ($m = 3^a q^b$, $q \geq 7$, $\max(a, b) > 1$).

- Proposition 7.4 ($m = p^a$), Theorem 7.3 and Theorem 7.8 → Theorem 1.3.
- Lemma 7.5 and (7.7a) → Lemmas 7.9 and 7.10 → Theorem 7.11 ($9 \mid m$, using Lemma 7.18 for non-decomposability at 45).
- Proposition 7.4, Theorem 7.3 (at $m = 15$), Theorem 7.11 and (7.7a) → Theorem 7.12 ($m = 3^a 5^b$).
- Lemma 7.5 → Lemma 7.14; Theorems 7.3, 7.6, 7.11, Lemma 7.14 and (7.7a) → Theorem 7.15 ($5 \nmid m$).
- Lemma 7.5 and Lemma 7.1 → Lemma 7.16 (short kernels prime to six).
- Lemma 7.10, Lemma 7.16 and (7.7a) → Lemma 7.17 (odd surface classification).
- Lemma 7.17 and Theorem 7.3 (at $m = 21$) → Lemma 7.18 (four points of a 5-progression).
- Theorem 7.6, Lemma 7.16 and (7.7a) → Lemma 7.19 (zero row difference $D = 0$).
- Proposition 7.4, Theorems 7.3, 7.6, 7.8, 7.11, 7.12, 7.15, Lemmas 7.16, 7.18, 7.19 and (7.7a) → Theorem 7.20 and Theorem A (compiled in Proposition 7.21; Theorem 1.4 combines Theorems 7.3, 7.8, 7.11, 7.12, 7.15 and Proposition 7.4).

At the eight degrees $3, 5, 15, 21, 33, 39, 45, 105$ reached by these arguments, Proposition 4.1 and a further, deliberately naive enumerator, which loops over all sorted sextuples with sum $3m$ and checks every unit without orbit normalization, list 28 pair-free orbits: 6 exactly 5-standard, 16 of the remaining orbits in Dec and the six seeds of (3.1), in agreement with the case analysis above (Appendix).

Data availability

The source code, data tables and verification output for the finite computations of Section 4 and the Appendix are contained in Supplement S1, the archive described in the Appendix. The proof of Theorem A does not depend on these computations.

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[17] T. Shioda and T. Katsura, *On Fermat varieties*, Tohoku Math. J. 31 (1979), 97–115. DOI: 10.2748/tmj/1178229881. Theorem 1.7 and Proposition 2.4 give the blowup–quotient construction underlying [16, Theorem II].

Appendix. Reproduction of the finite arithmetic

Supplement S1 is the accompanying archive `calculations-bundles.zip`, identified by the supplied `CONTENTS.sha256` file (together with the top-level README file in the supplement). It contains the complete source and data used below, including both implementations of the census through 105, their raw outputs and the larger census outputs. It can be extracted and replayed without this project's repository, remote machines or pre-existing executables. Its README file gives the commands and scope; `CONTENTS.sha256` identifies `calculations-bundles.zip` and every enclosed source and data file. All bundle names below are paths under `calculations/bundles/`, which are inside `calculations-bundles.zip` after extraction. The archive is supplied with this local manuscript; no public deposit or DOI is assumed.

The geometric arithmetic and calibration bundle is `odd-fourfolds-le599-checked-closure`.

The additional checker is

`verify_short_semiprime_classification.py`

It prints the complete `short-semiprime-raw.jsonl`: twelve finite calibration cases, including all four small semiprime bases of Theorem 7.3, checked by direct age equations and explicit surface or standard witnesses. The unbounded classifications follow from the proofs in Section 7.

An additional independent checker and its complete output are

`verify_two_prime_depth.py`
`two-prime-depth-check.jsonl`

They check the short split matrices in Lemma 7.7 and sparse potentials in Theorem 7.8, using direct age equations and surface splits at degrees 147, 363, 507, 867, 1029 and 1083. These are calibration degrees, not an enlarged exhaustive census bound. Theorem 7.8 is proved analytically.

The additional finite checks for Lemma 7.9 and Theorem 7.12 are `verify_depth_extensions.py` and `depth-extensions-output.jsonl` in the same bundle. They classify all 66, 36 and 6 short matrices without a same-sign three-line at sizes 3×4, 3×5, 3×6, and check the nonzero row-difference branch and all normalized sparse potentials at degrees 75, 375 and 1875.

The bundle `quota-queue-r3-independent-enum-2026-10-29` supplies the unchanged C enumerator, its direct Python checker, the extension driver `extend_residual_census.py`, and the complete `extension-output.txt`. The extension checks 103 selected degrees above 599, together with the original controls: 156 distinct degrees, 966 nonpair Hodge orbits and 90 residual orbits. The selection uses only the previously established families, so is independent of Theorems 7.11, 7.12 and 7.15. Run on one CPU with a C11 compiler and Python 3:

```
python3 extend_residual_census.py > replay-extension-output.txt 2>&1
```

The driver checks every output orbit with all units, all three-three splits and exact standard multisets. It also reruns the original small direct census controls. No bounded test proves the unbounded assertions.

`verify_positive_finite_reduction.py` imports no census or earlier checker. Its complete output is `positive-finite-reduction-output.jsonl`. It directly enumerates the primitive pair-free surface orbits at 26 calibration degrees from 5 through 1155. The only non-3-standard ones occur at 15 and 21, as Lemma 7.17 requires. It also checks all sixteen four-sign Fourier patterns using exact Gaussian integers, and all thirty-two five-sign polynomials by reduction modulo $1+X+X^2+X^3+X^4$. These finite checks are independent of the analytic conductor bound. Proposition 4.1, replayed independently as described below, is a check of the degrees through 105; Theorem 7.20 does not use it.

The separate bundle `xcheck-hodge-thm720-2026-10-29` contains the independently written `hs_enum3.c`. It uses a three-plus-three enumeration: fix a normalized entry at a proper divisor of m, enumerate the other two entries of its triple, and match against all sorted triples with the complementary sum and age sums. The matching key tests at most 32 units; every candidate is then checked at all units, using the symmetry between t and $-t$. This key is only a filter and cannot remove a balanced tuple. Unit normalization is justified by the argument in Section 4, and all possible triples are included. Thus this is an independent exhaustive implementation, not a test of only the predicted residuals.

Its `results/r4-20261030/` retains a fresh run at every odd degree from 3 through 105. The two additional calibration degrees $3465 = 3^2 \cdot 5 \cdot 7 \cdot 11$ and $3675 = 3 \cdot 5^2 \cdot 7^2$ use the separate pair-plus-triple enumerator from `quota-queue-r3-independent-enum-2026-10-29`, with only its input bound increased to 3675. Its source snapshot is `residual_pair_triple_through3675.c`. The script `verify_r4_census.py` checks completion, all returned residuals by direct full-unit age and closure predicates, and equality with the predicted seed-inflation sets. The larger two degrees are individual checks; they do not extend the continuous census bound 2001 or prove the unrestricted classification.

The review bundle `xcheck-hodge-r5-2026-11-02` supplies a second implementation of the census of Proposition 4.1. Its `enum6.c` and `classify.py` are the independent reviewers' original code. The three-plus-three source `hs_enum3.c` in `xcheck-hodge-thm720-2026-10-29` has only an added optional output line per classified orbit; its enumeration and predicates are unchanged. From the extracted Supplement S1 root (after extracting `calculations-bundles.zip`), run on one CPU with a C compiler and Python 3:

```
python3 -I calculations/bundles/xcheck-hodge-r5-2026-11-02/replay_base.py \
    scratch/finite-base-replay
```

The output directory must be new. The driver builds both executables, retains all raw rows and compiler diagnostics, checks every returned direct-enumerator tuple at every unit, and compares every canonical pair-free orbit and its category with a fresh three-plus-three run. Neither enumeration reads archived answers or the seed list. The seed list is used only after enumeration for comparison with the resulting residual set. A nonzero compiler or enumerator exit, missing degree, or classification disagreement fails the replay. The retained complete replay is `results/r8-20261102/` in the direct-enumerator bundle; its final counts are 52 degrees, 57 pair-free orbits, seven residual orbits and six primitive residuals. On macOS, an SDK may be selected with `CFLAGS='-O2 -isysroot /path/to/MacOSX.sdk'`; on other systems the default `cc -O2` suffices.

The bundle `r9-large-degree-mitm-2026-11-04` contains a third, separately written enumerator, `mitm6.c`. It matches two triples, one containing a proper divisor of m, by a hash of the sum and of the ages at sixteen fixed units, and then verifies every match at all units. With `classify_r9.py` it reproduces the 57 pair-free orbits of Proposition 4.1 with the same categories. It was also run at the ten degrees 1155, 1365, 1785, 1995, 2145, 2415, 2625, 2835, 3003 and 3315. At each of them the residual set consists exactly of the seed inflations predicted by Theorem A, and the residual counts agree with the formula for $|\mathcal{R}(m)|$. The residual sets and category counts also agree with two earlier, separately written enumerators wherever these were run; verbatim copies of their outputs are in the bundle's `prior-runs/`. These ten degrees are observations only; no proof in this paper uses them. The bundle README file gives the commands and the retained raw output.

The bundle `r10-minimal-base-2026-11-04` checks Proposition 4.1 at the eight degrees of Proposition 7.21 with `naive6.c`. It loops over every sorted sextuple of nonzero residues with sum $3m$, forcing the last entry, rejects inverse pairs and tests every unit; it uses no orbit or divisor normalization and no hashing. Its rows are classified by

`classify_r9.py`. The canonical orbit sets agree with those of the other implementations at all eight degrees (28 orbits; Proposition 7.21). The run was a single batch job; raw rows and logs are retained in the bundle.

The bundle `xc-uniform-check-2026-11-04` contains `xc_uniform_check.py`, written separately from all of the above, and its reference output. It enumerates the pair-free Hodge sextuples at the same eight degrees by matching unit age vectors of triples, and reproduces the 28 orbits and their categories. It also lists all length-six solutions of the potential equations (7.6a) at 15, 21, 33 and 39, all sextuples containing four points of a 5-progression at 45 and 105, the absence of balanced configurations in the branches of Theorem 7.20 at 105, and, at 45 and 105, that every pair-free surface quadruple with two entries differing by an element of order five is killed by 15. These are checks of the case analysis in Sections 7.1, 7.5 and 7.8, not inputs to it.

The bundle `xcheck-hodge-thm720-2026-10-29` also checks the explicit contraction (7.14e) against the lower-stratum construction. Run:

```
python3 block_E0.py 35
```

This checks all surviving signed potentials of half-norm at most three at the eleven moduli prime to six from 5 through 35. The output records 190 surface quadruples, 190 sextuples with the displayed contraction, and four invariant blocks at $N = 13$. These counts refer to potentials, not unit orbits; the general proof is Lemma 7.19. The complete output is retained in `results/r5-20261102/`.

All bundles are supplied under `calculations/bundles/` inside `calculations-bundles.zip` after extraction. To replay the original census, enter `odd-fourfolds-le599-checked-closure` and compile:

```
gcc -O3 -Wall -Wextra primitive_residual.c \
    -o /tmp/hodge-primitive-residual
```

Run the executable once for each odd $3 \le m \le 599$ divisible by 3 or 5, in increasing degree order, concatenating standard output to `primitive-census.txt`. The other degrees are covered by [1, Theorem A]. Then run `python3 verify_closure.py` from the bundle directory. The Python checker independently verifies the six seed identities, all surface and curve age equations, the non-coprime standard reductions, both gap detectors, the list of completed degrees, and the 38 inflated residuals. It also checks Proposition 6.1's family for every odd degree from 5 through 2001, and all half-sign potential models in the six stated semiprime calibration cases. Its scope is exact finite arithmetic; the geometric and unbounded assertions are proved in the text.